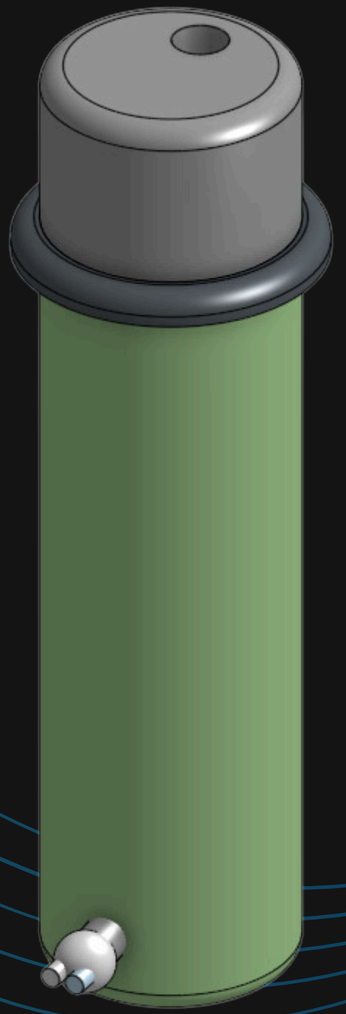
# Thep Hackathon SS2

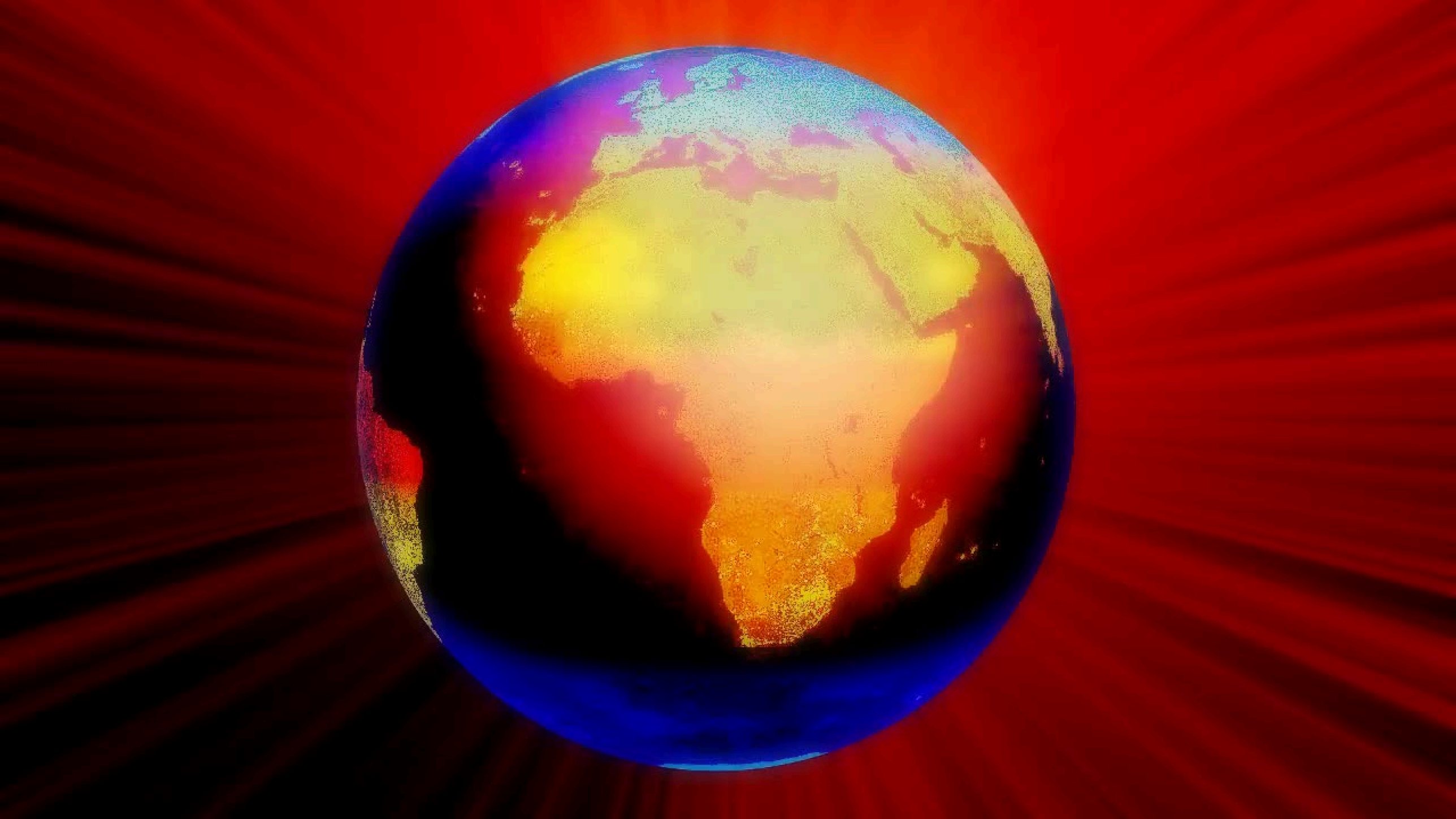
# Algal Carbon Capture : Bio-Based solution for solving Climate Change



FINAL

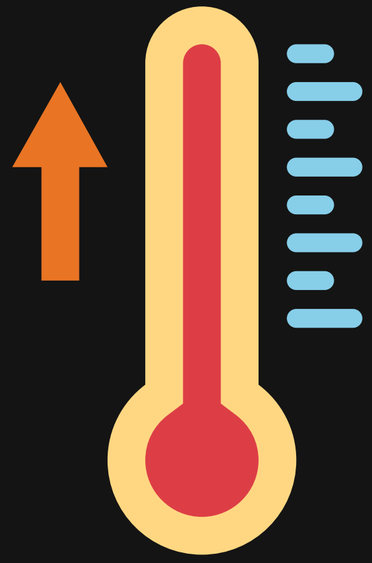
MATTHEW

PEEMPONG



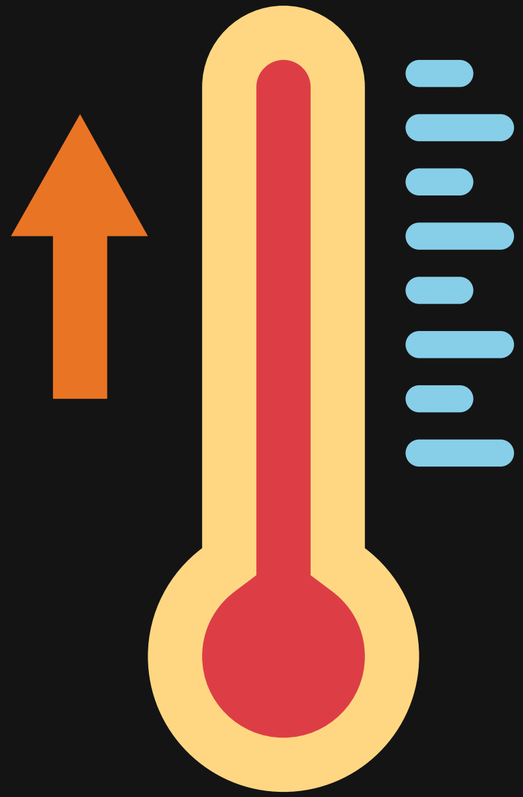
# Introduction

19TH



TEMPERATURE INCREASED BY
0.18°C PER YEAR

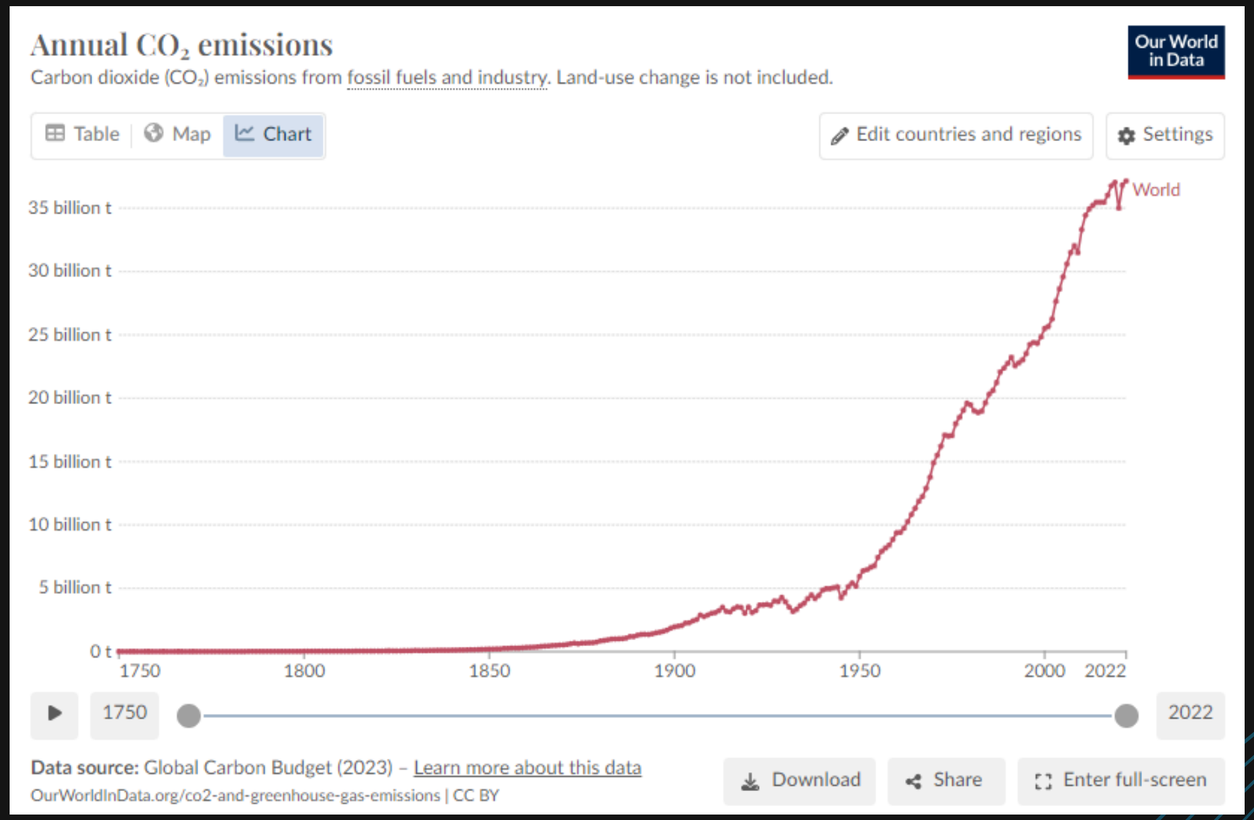
20TH

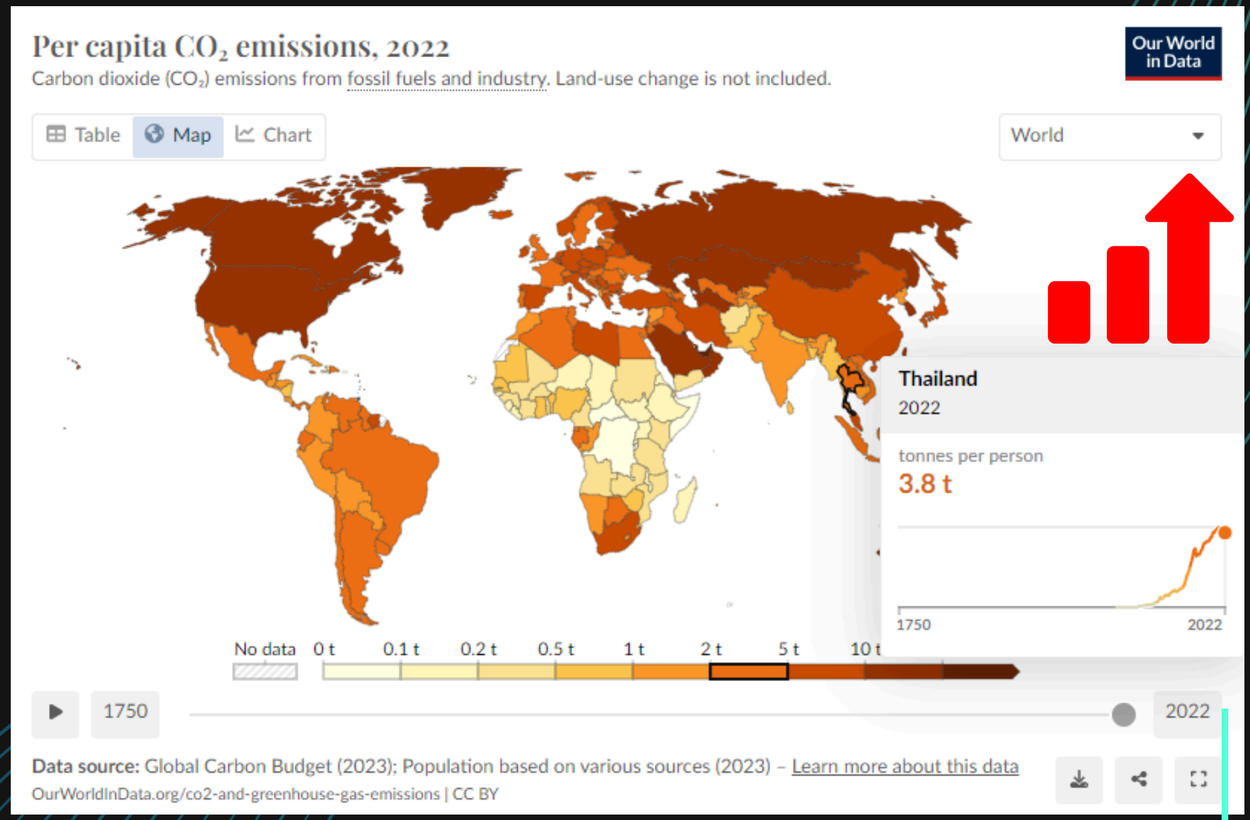


TEMPERATURE INCREASED BY
0.74°C PER YEAR

CLIMATE CHANGE

# Introduction



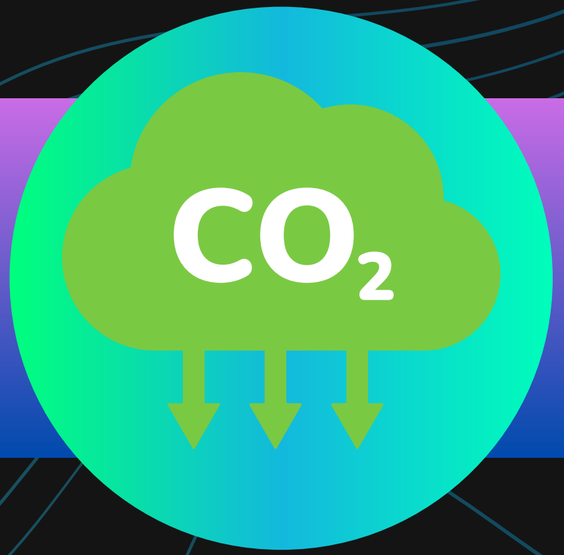

(https://ourworldindata.org/co2-emissions, 2024)

# Challenge

Energy Conservation

Clean Enerygy

Plant Trees




Remove CO2

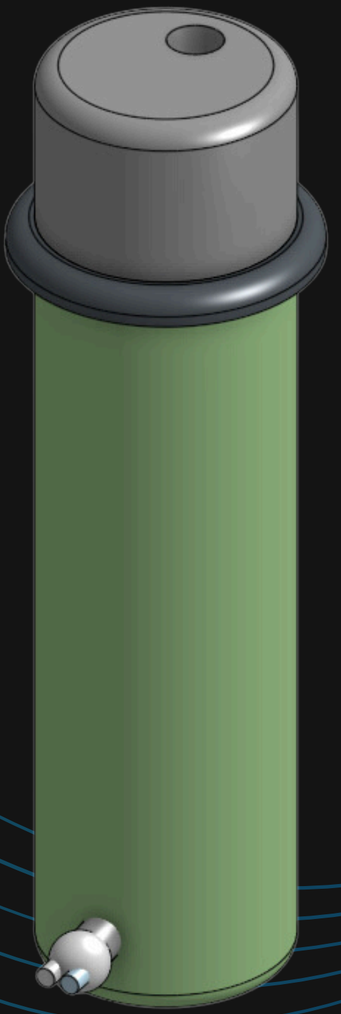
# Target

Remove CO2

**1.3 million tonnes of CO2** emissions might be avoided if **1%** of the global population could **absorb 16 kg** of CO2 annually.

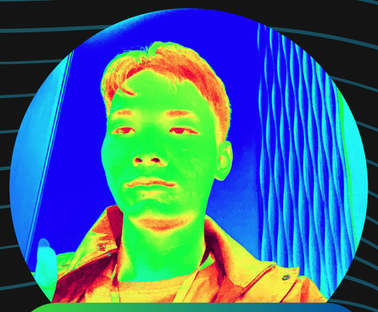
# Thep Hackathon SS2

# Algal Carbon Capture : Bio-Based solution to tackling Climate Change



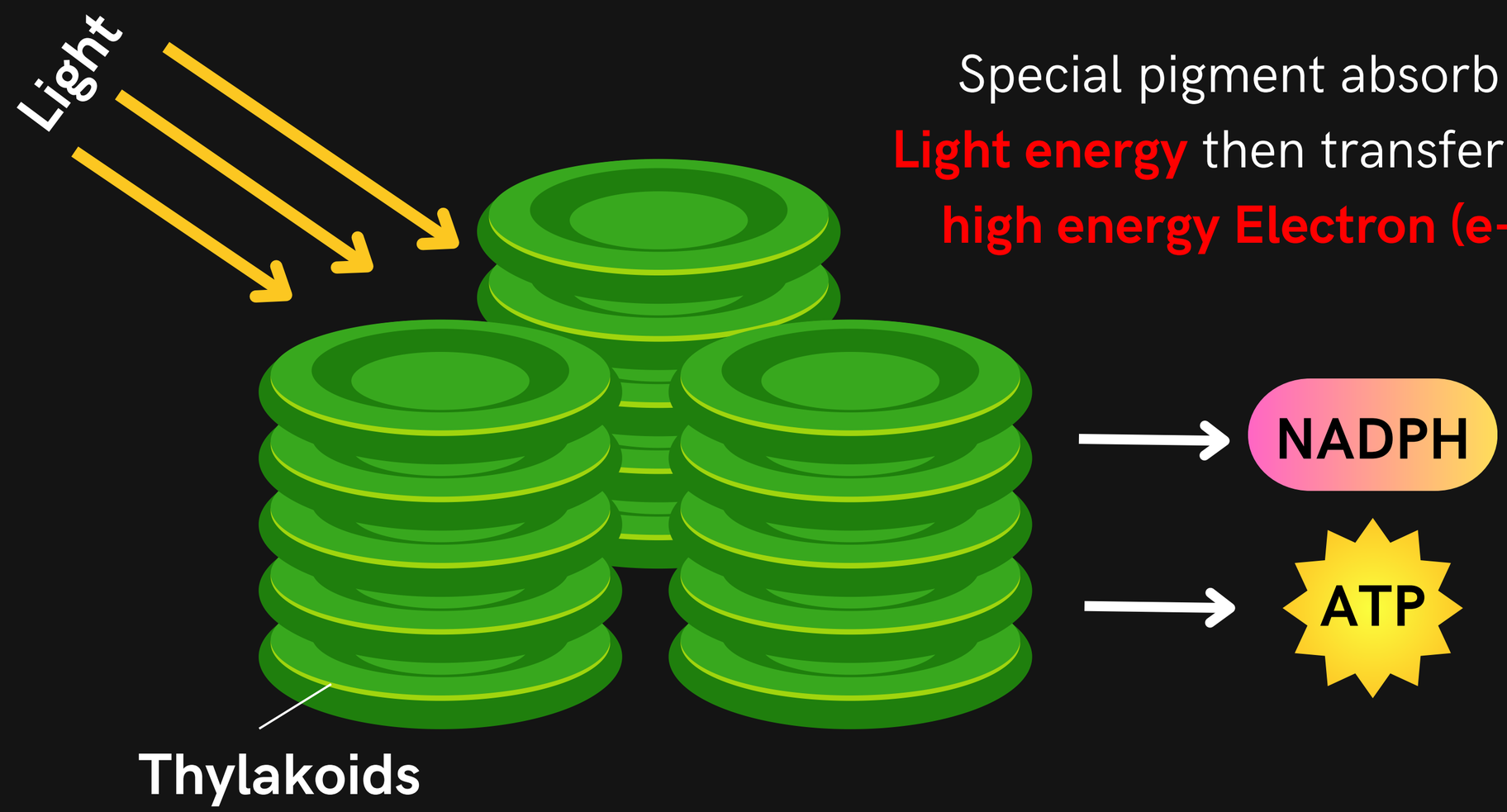
FINAL

MATTHEW



PEEMPONG

# The Light Reaction of Photosynthesis

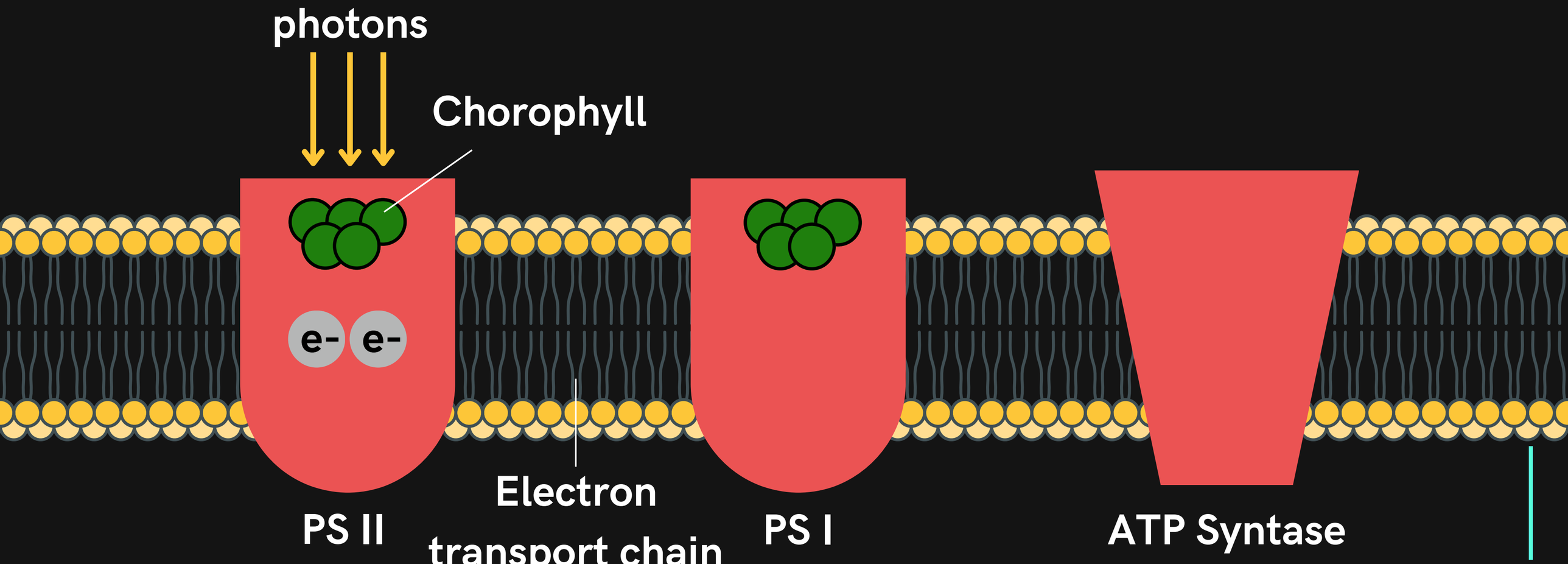



Special pigment absorb
**Light energy** then transfer to
**high energy Electron (e-)**

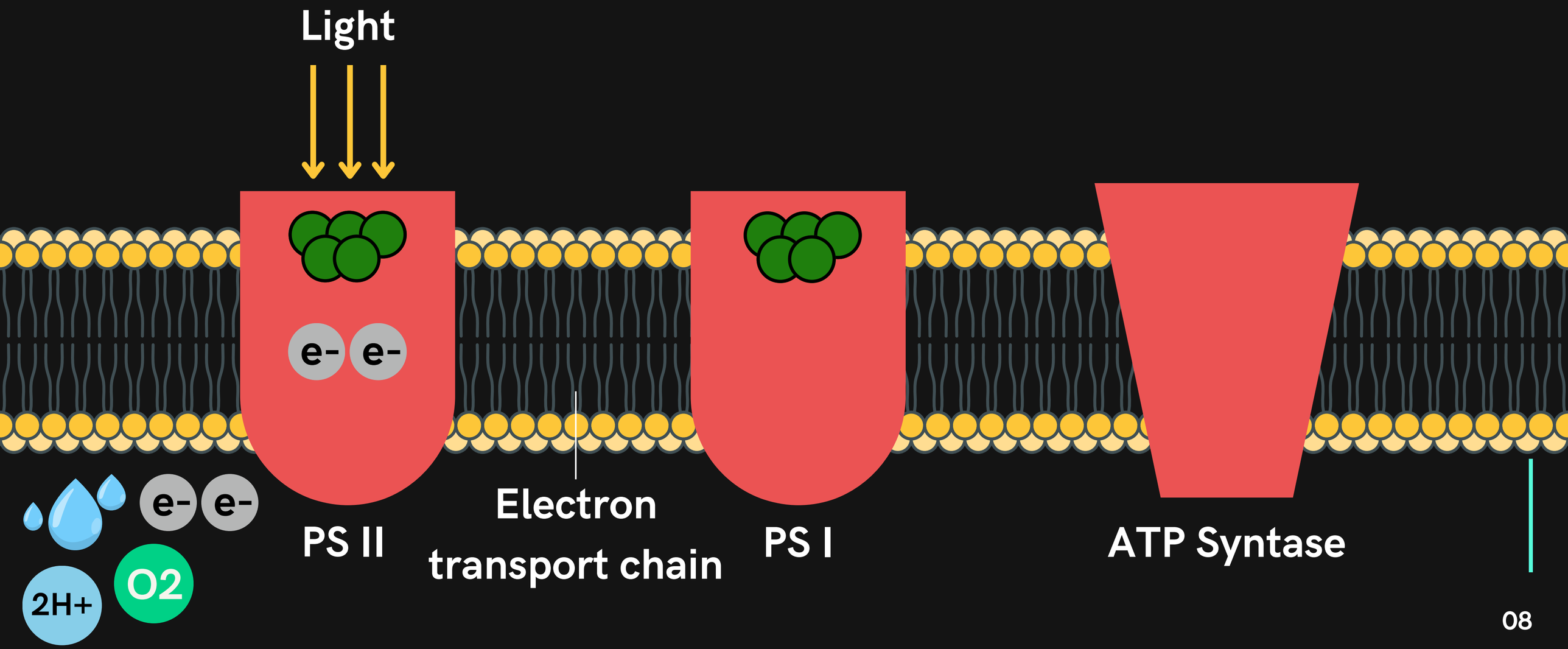
NADPH

ATP

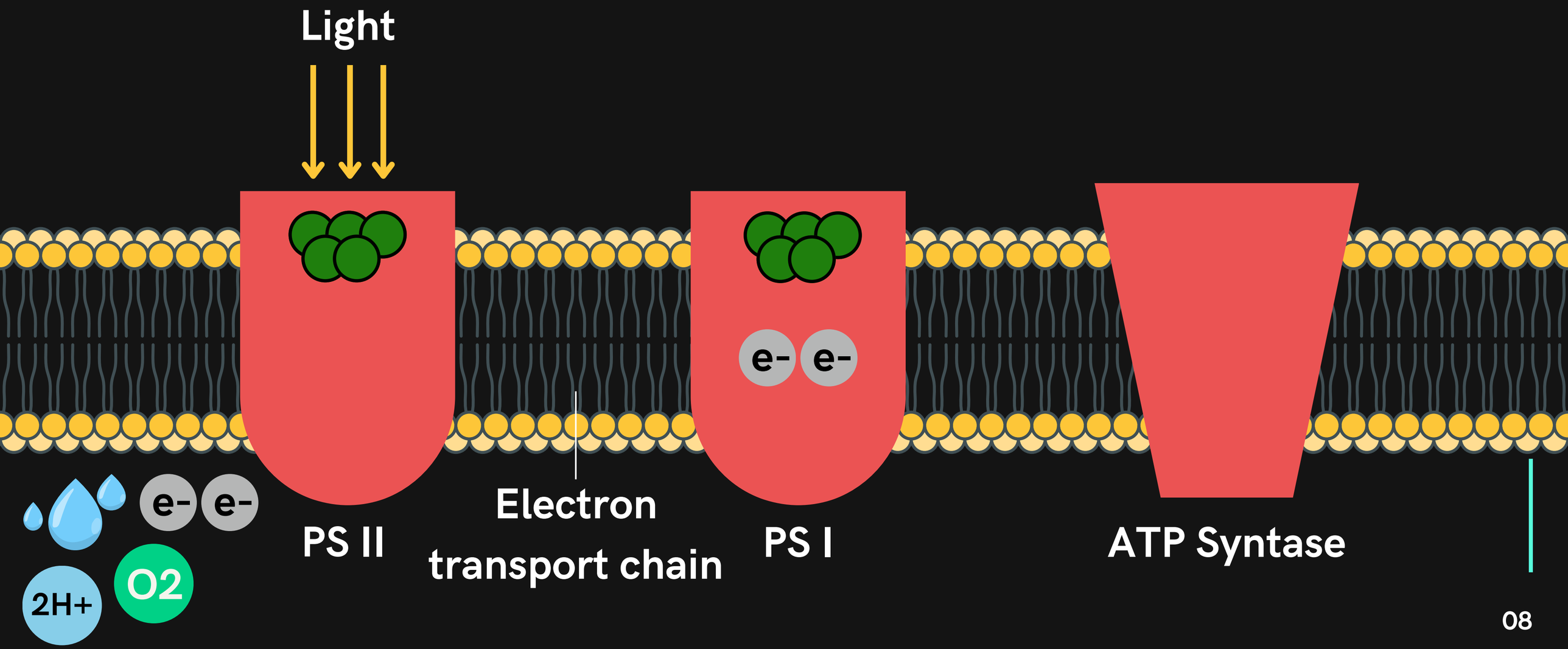
# The Light Reaction of Photosynthesis



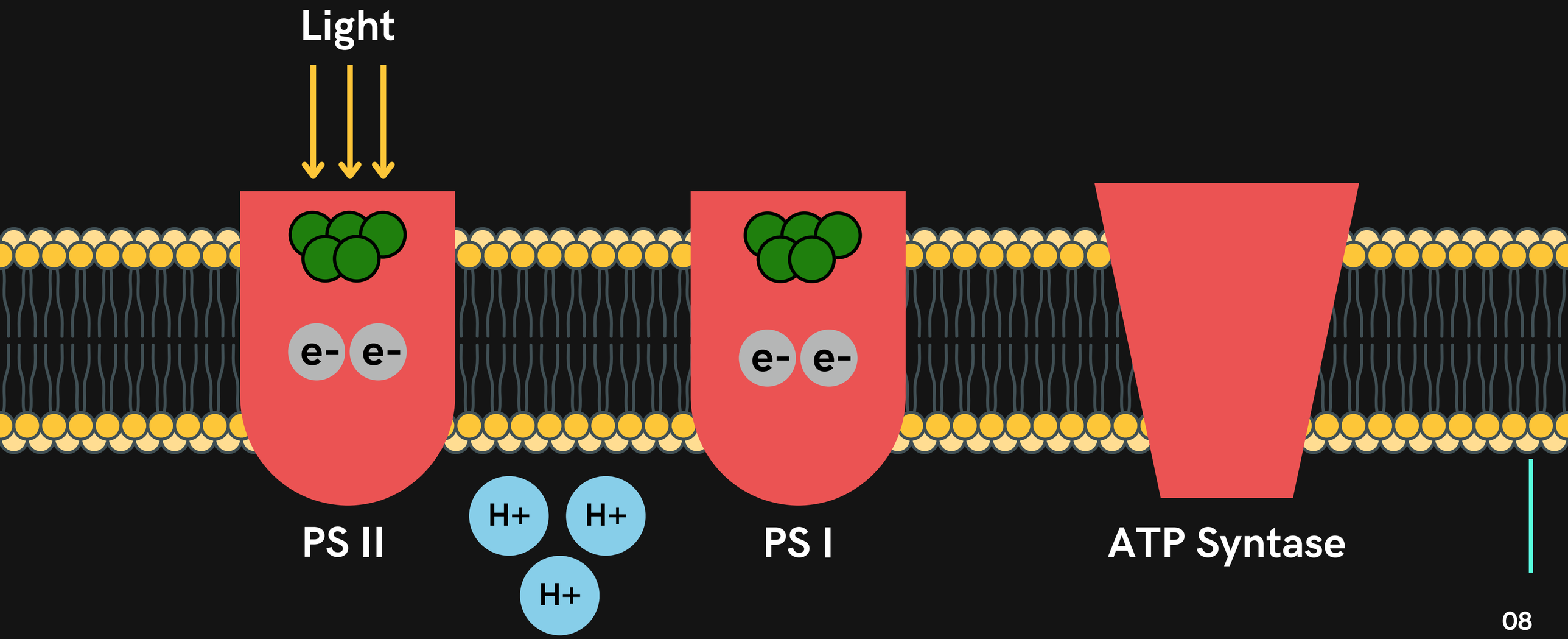
# The Light Reaction of Photosynthesis



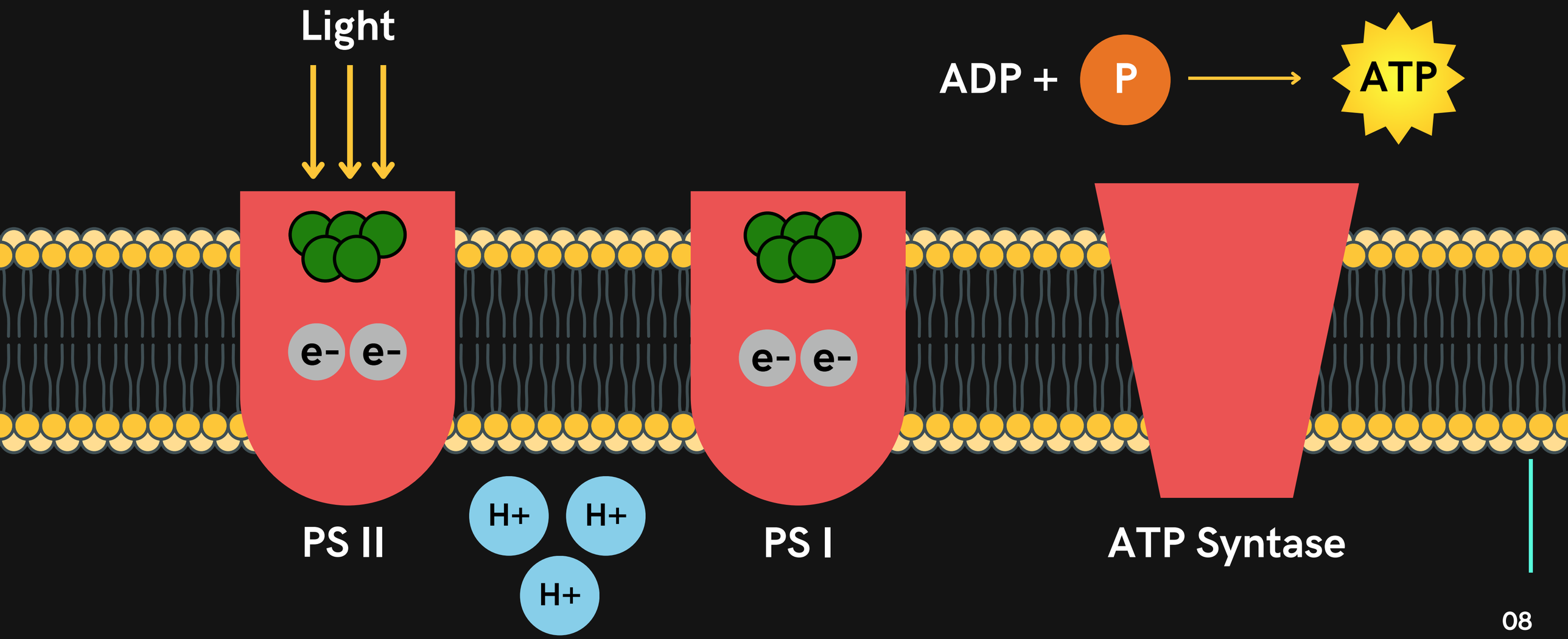
# The Light Reaction of Photosynthesis



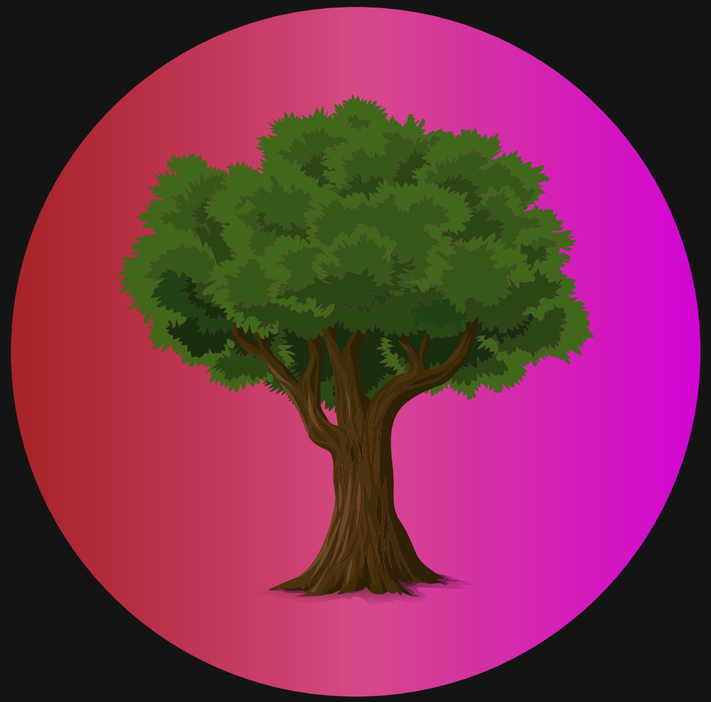
# The Light Reaction of Photosynthesis



# The Light Reaction of Photosynthesis



# Photosynthesis



Plant Trees

## A tree absorbs about 10 kg of $CO_2$ anually

$$CO_2 + H_2O \xrightarrow{\text{light,chlorophyll}} C(H_2O) + O_2 + 120\ \text{kcal/mol}$$

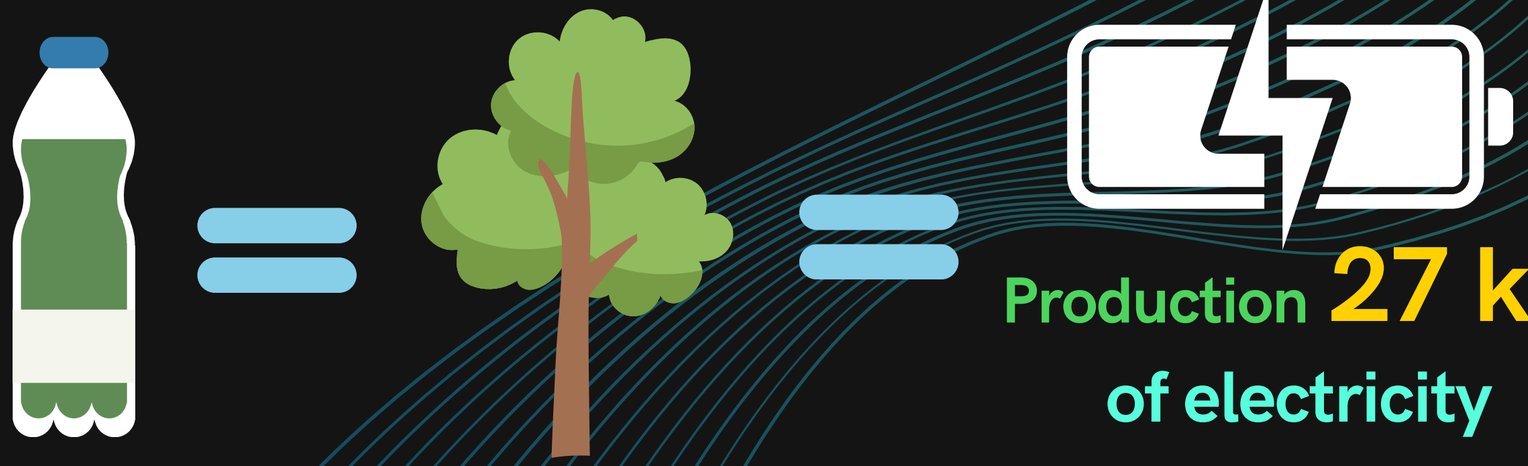
Stephenson, N., Das, A., Condit, R. et al. Rate of tree carbon accumulation increases continuously with tree size. Nature 507, 90–93 (2014). https://doi.org/10.1038/nature12914

# Photosynthesis

**Algae**

*CHLORELLA SP.*

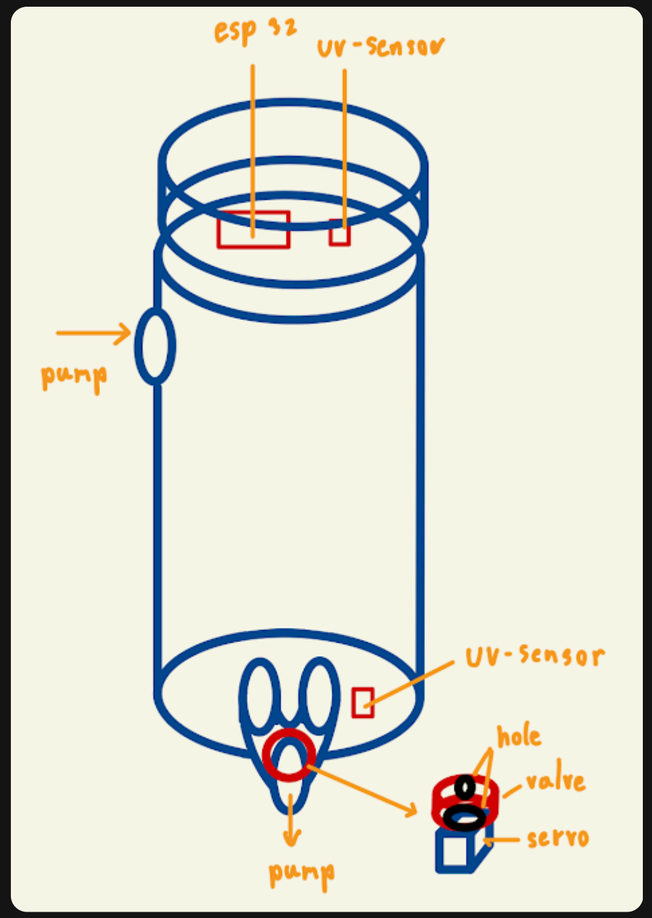
## This 2L of Algae able to absorb 16 kg of $CO_2$



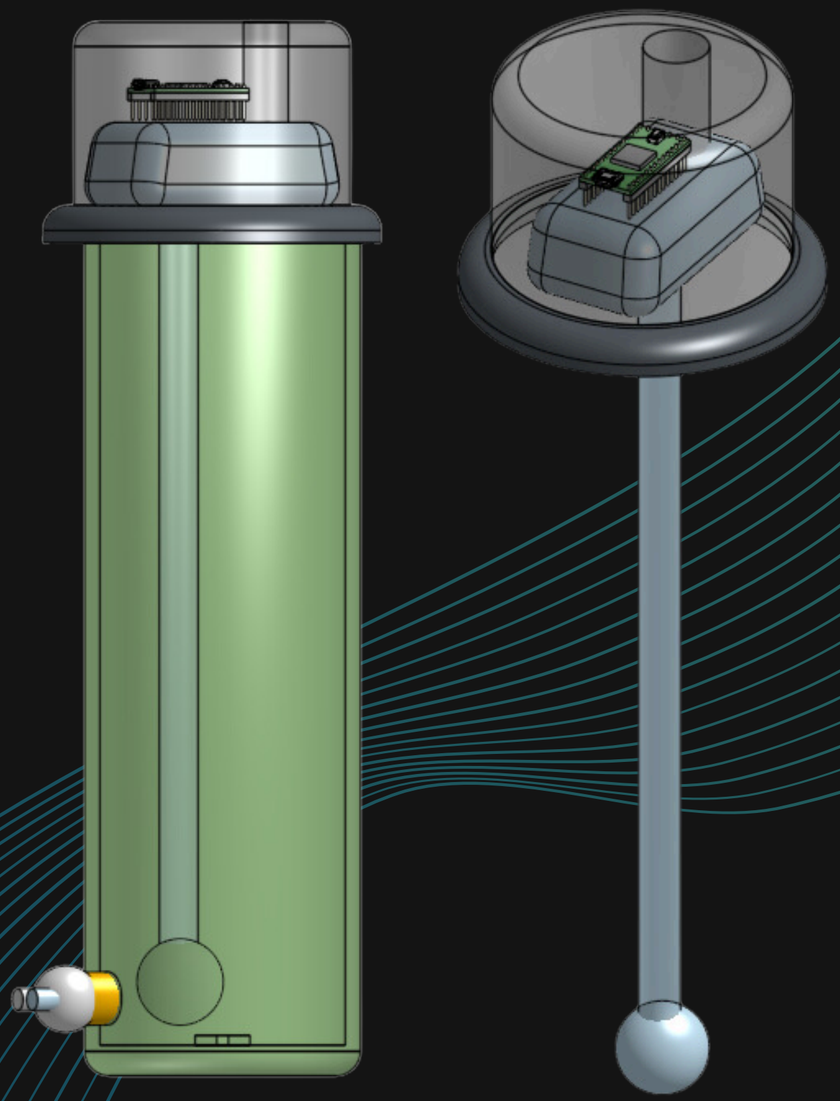
Production **27 kw** of electricity

Ammann, E. C. B., & Fraser-Smith, A. (1968). Gas exchange of algae. Applied Microbiology, 16(5), 669–672. https://doi.org/10.1128/am.16.5.669-672.1968

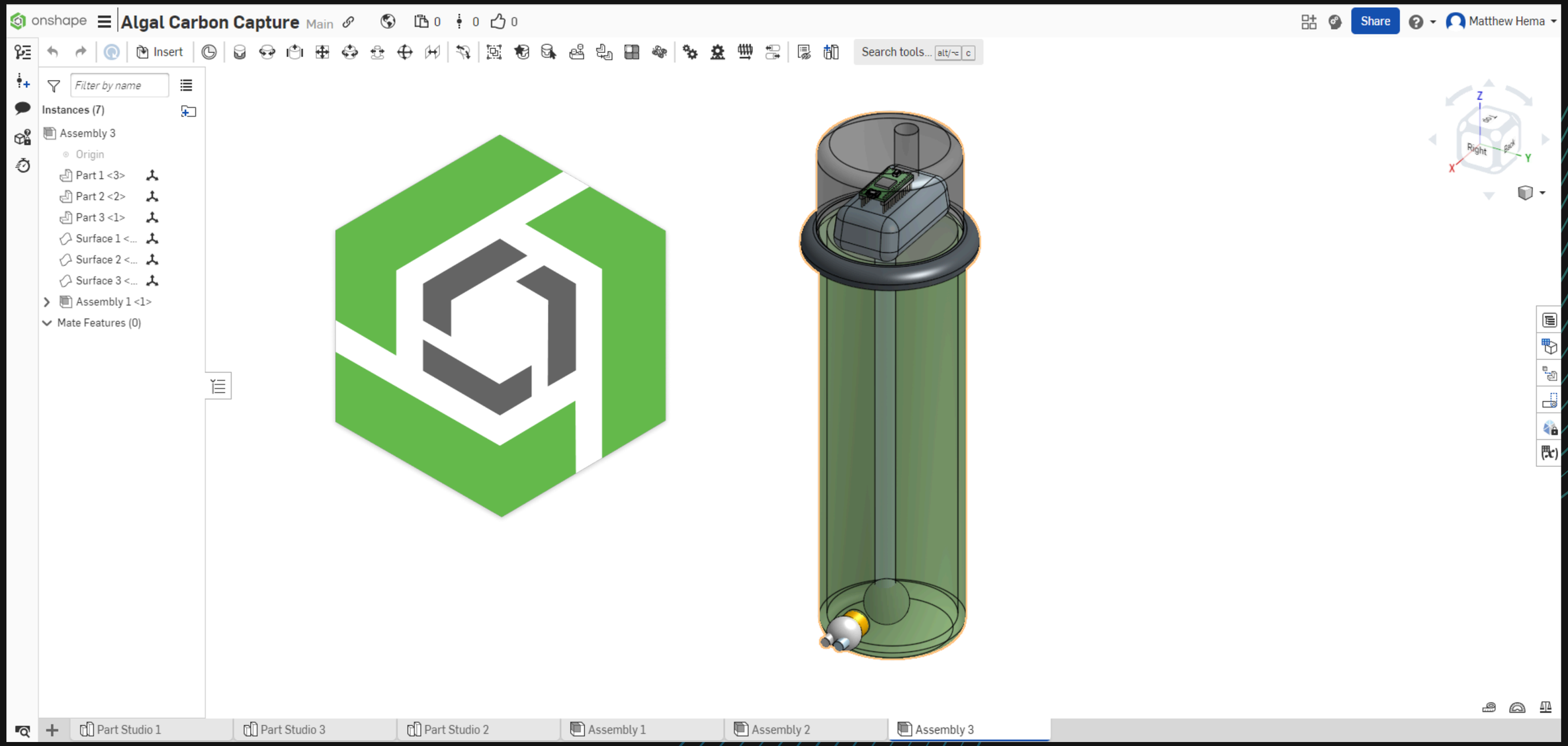
# Our Solution



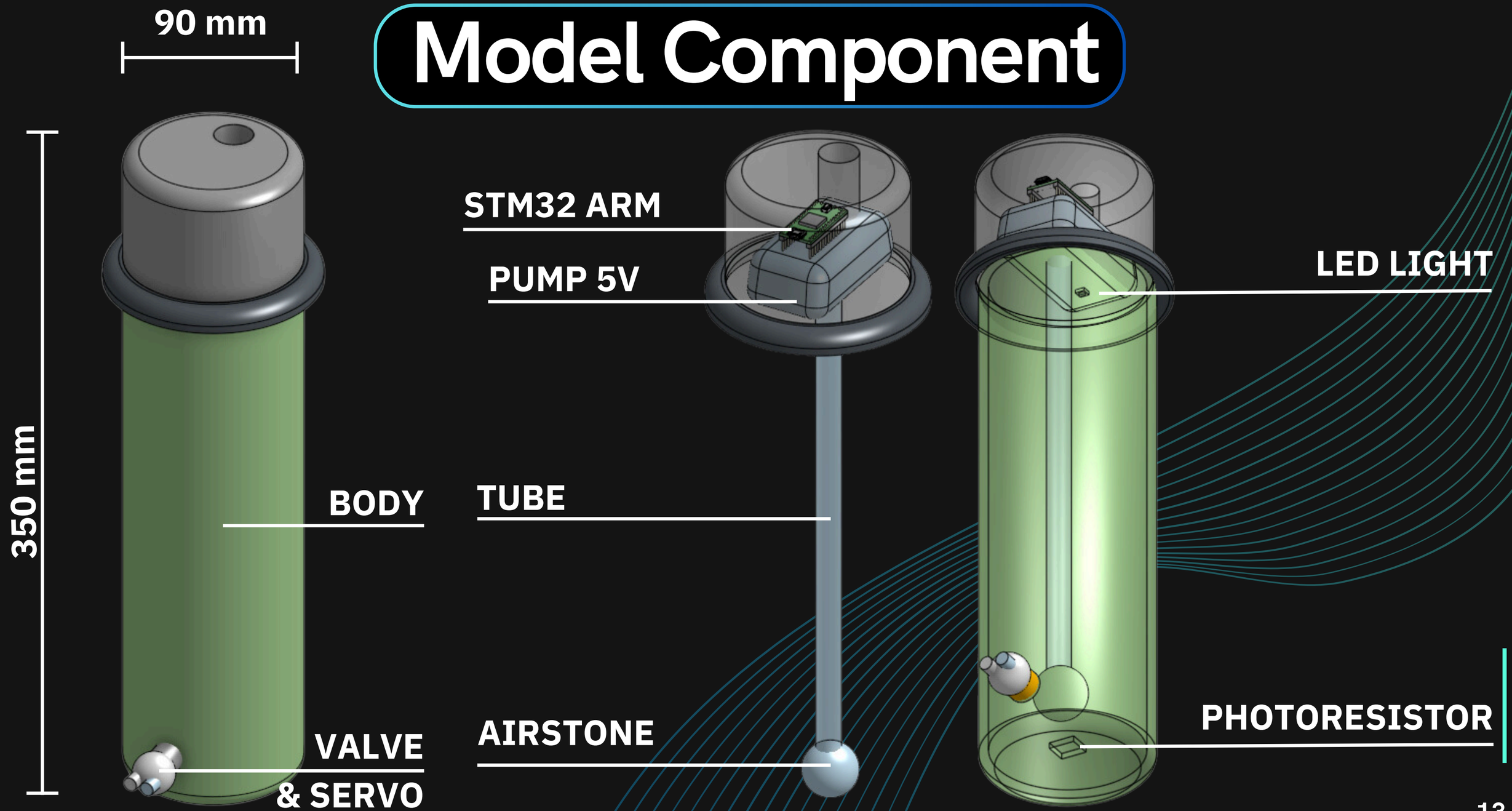

Sketch up



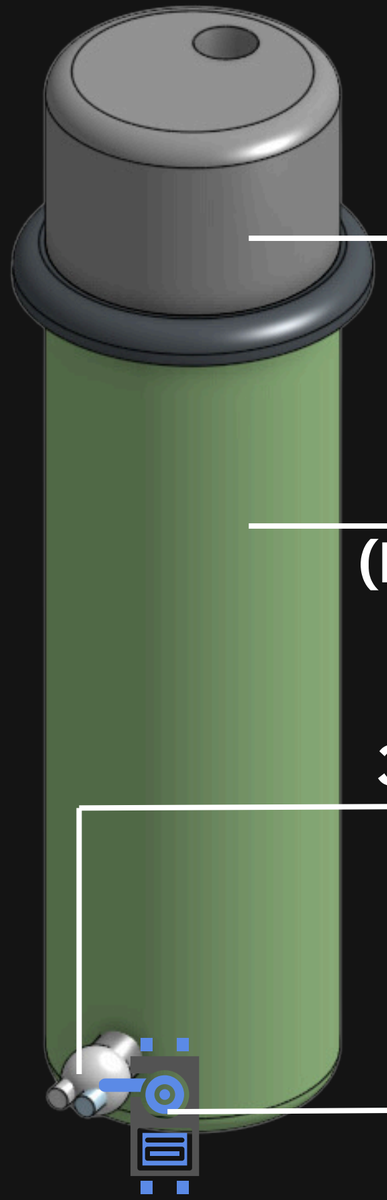
The Model


onshape
Algal Carbon Capture
Main
Share
Matthew Hema
Insert
Search tools...
Filter by name
Instances (7)
Assembly 3
Origin
Part 1 <3>
Part 2 <2>
Part 3 <1>
Surface 1 <...
Surface 2 <...
Surface 3 <...
Assembly 1 <1>
Mate Features (0)
Right
Part Studio 1
Part Studio 3
Part Studio 2
Assembly 1
Assembly 2
Assembly 3

# Model Component



# Model Component

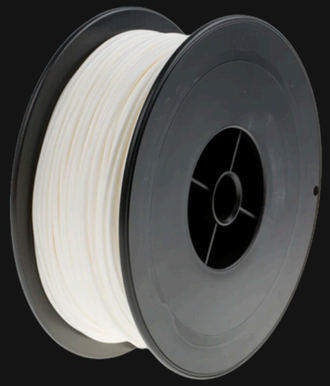


CAP
(3D Printing)

BODY
(Plastic Bottle)

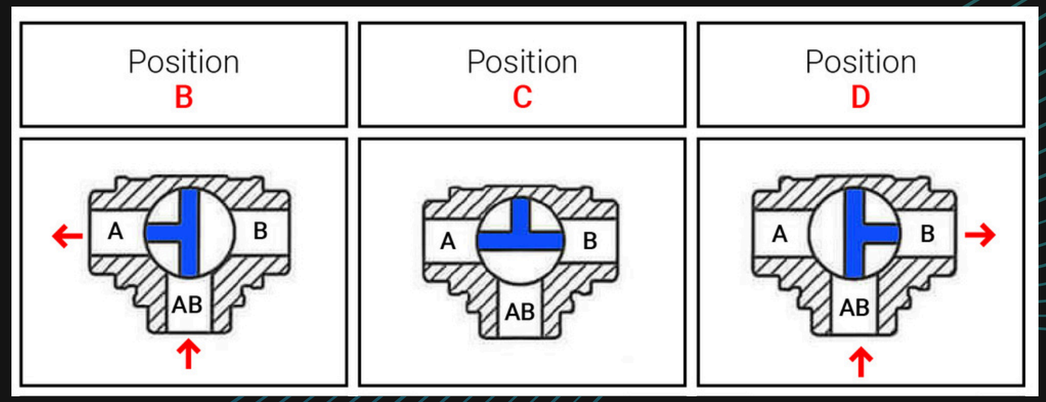
3-WAY VALVE
(3D Printing)

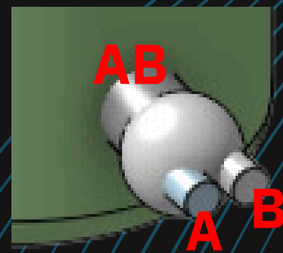
SERVO



**PLA filament**

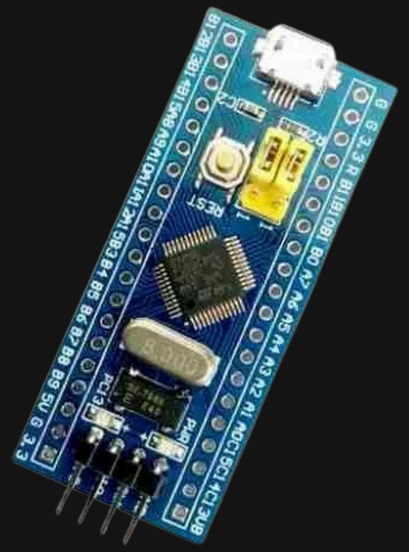
- Cheap
- Rapid



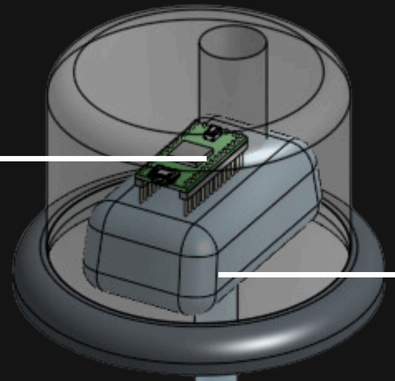




- A: Algae outlet (blank tube)
- B: Water outlet (5μm filter on)

# Model Component

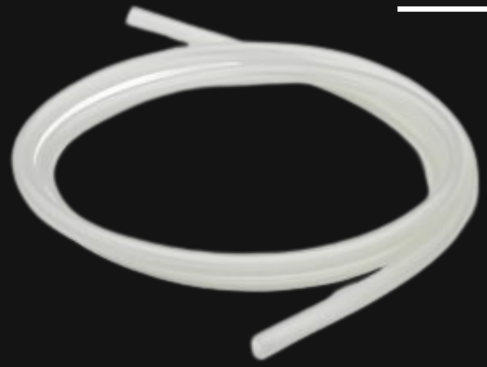


STM32 ARM



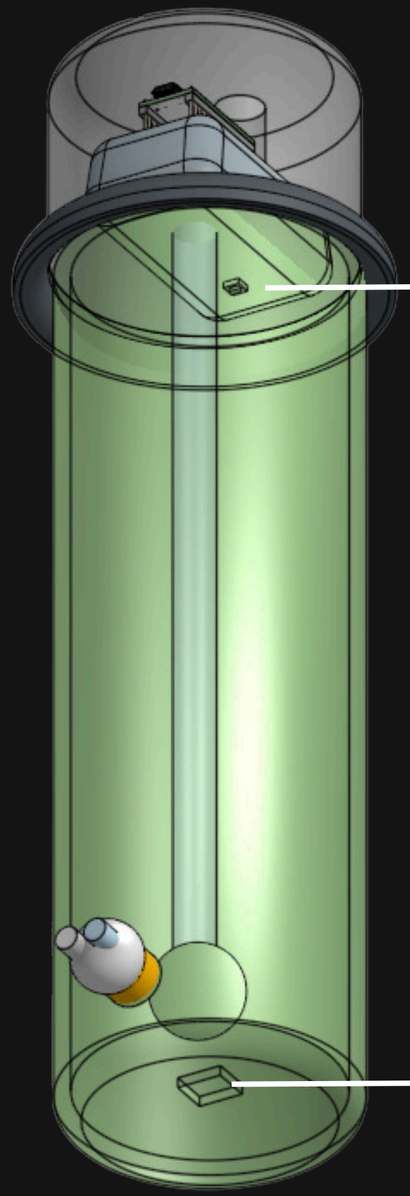
PUMP 5V

TUBE



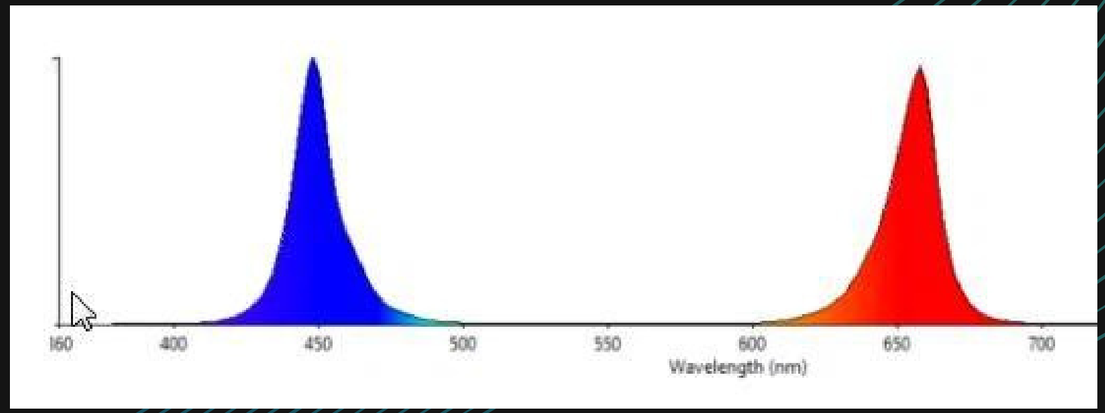
AIRSTONE

# Model Component

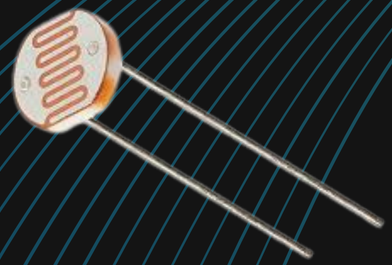


LED LIGHT

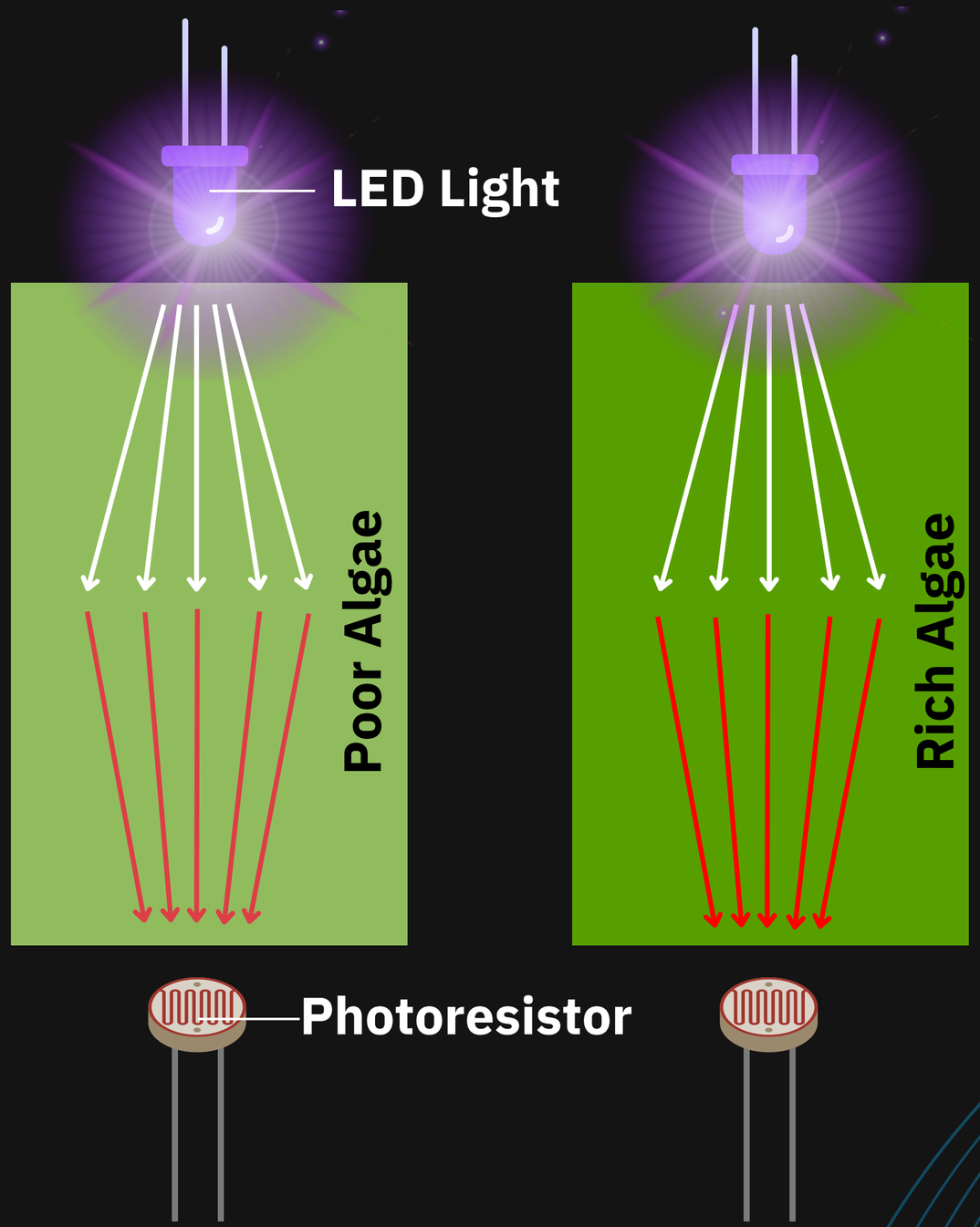
- For Calculating algae population
- Magenta color for Easier Calibrate




PHOTORESISTOR



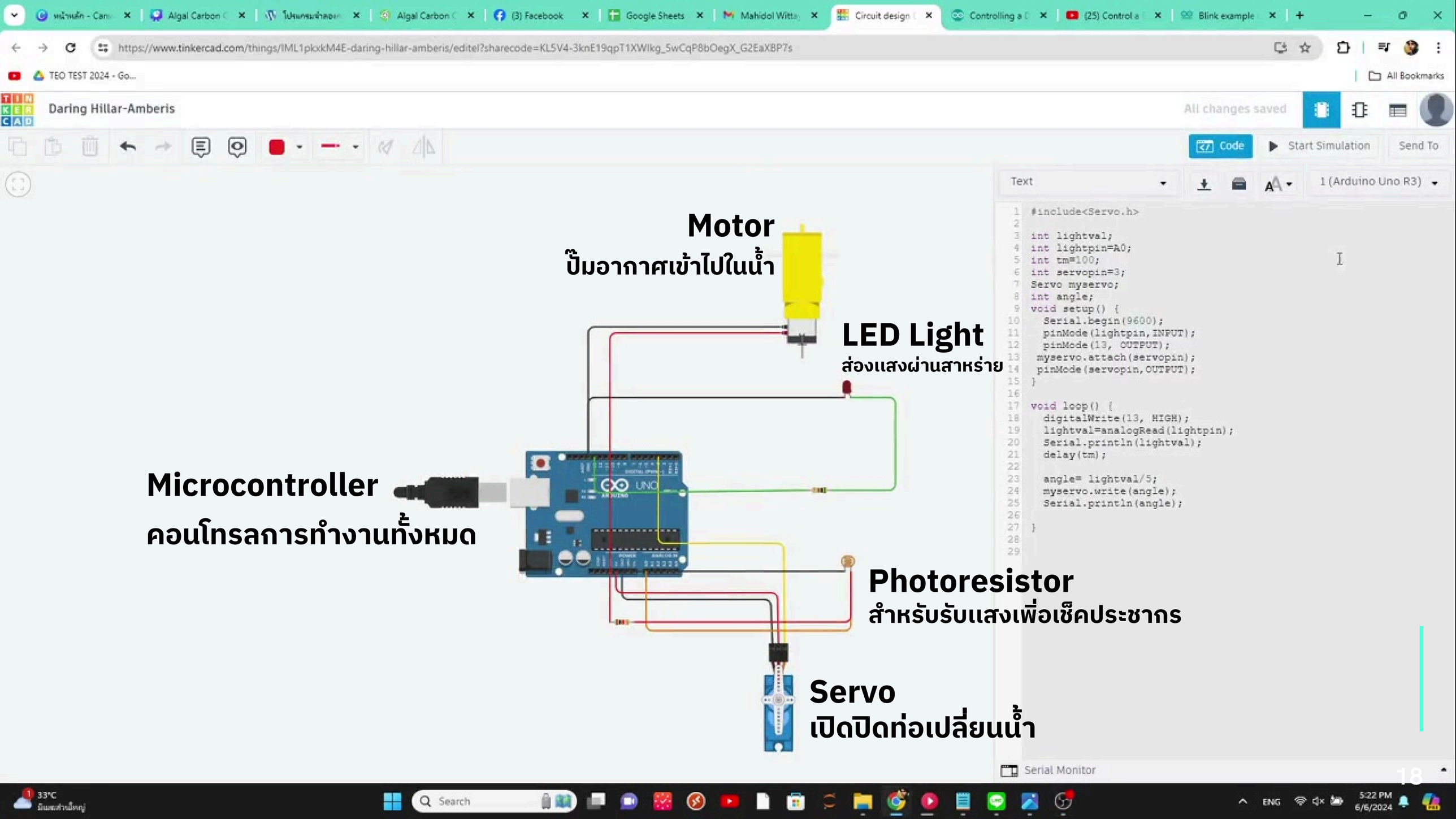
# Algae Population




$$\text{Transmittance} = \frac{I}{Io} \quad \begin{matrix}\text{(Output)} \\ \text{(Input)}\end{matrix}$$

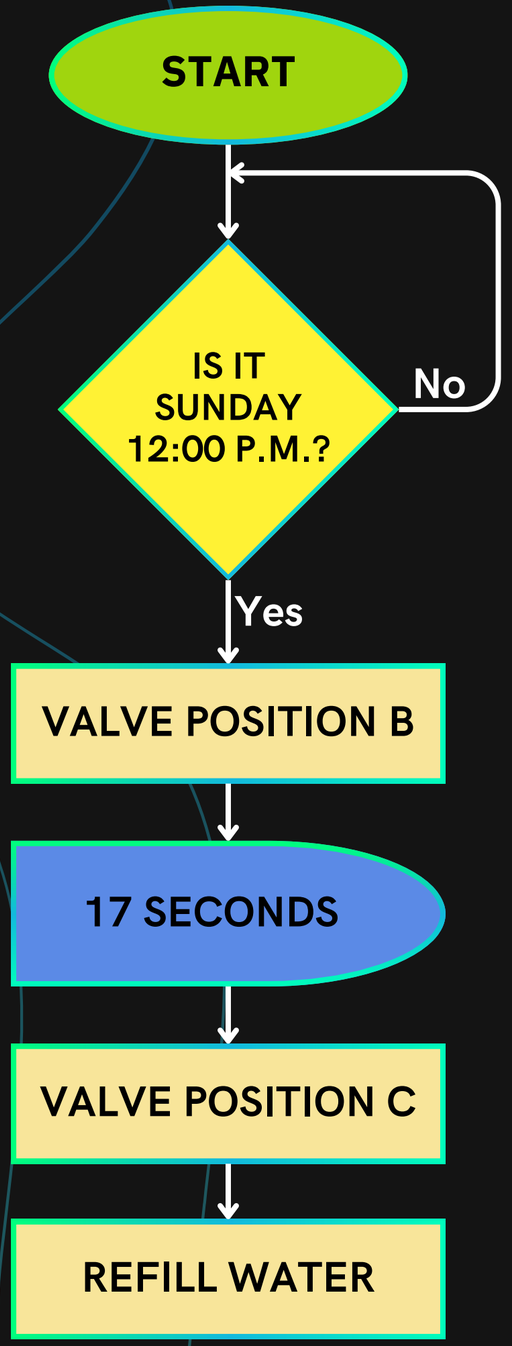
Input = water

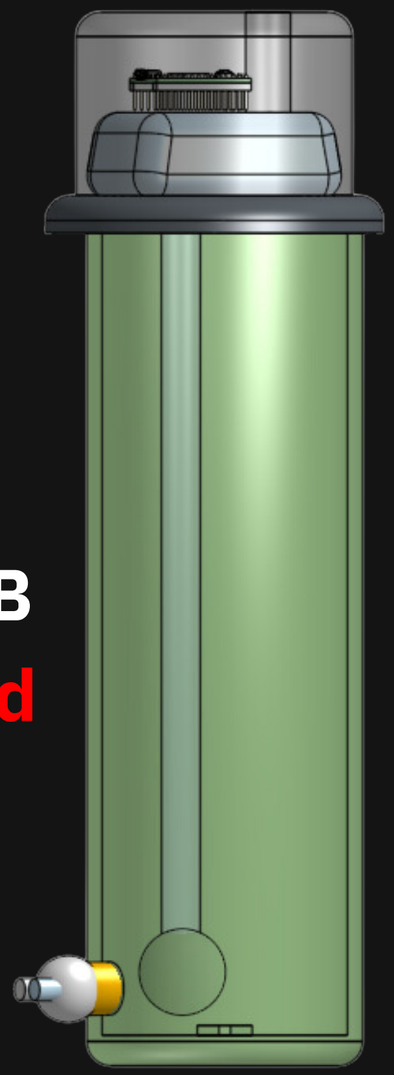
| Algae Population | Transmittance |
|---|---|
| Low | |
| Meduim | |
| High | |
| | |
| | |
| | |


Daring Hillar-Amberis
All changes saved
Code
Start Simulation
Send To
Text
1 (Arduino Uno R3)
#include<Servo.h>
int lightval;
int lightpin=A0;
int tm=100;
int servopin=3;
Servo myservo;
int angle;
void setup() {
Serial.begin(9600);
pinMode(lightpin,INPUT);
pinMode(13, OUTPUT);
myservo.attach(servopin);
pinMode(servopin,OUTPUT);
}
void loop() {
digitalWrite(13, HIGH);
lightval=analogRead(lightpin);
Serial.println(lightval);
delay(tm);
angle= lightval/5;
myservo.write(angle);
Serial.println(angle);
}
Motor
ป๊มอากาศเข้าไปในน้ำ
LED Light
ส่องแสงผ่านสาหร่าย
Microcontroller
คอนโทรลการทำงานทั้งหมด
Photoresistor
สำหรับรับแสงเพื่อเช็คประชากร
Servo
เปิดปิดท่อเปลี่ยนน้ำ
Serial Monitor

# Device Procedures

# Water Refill System

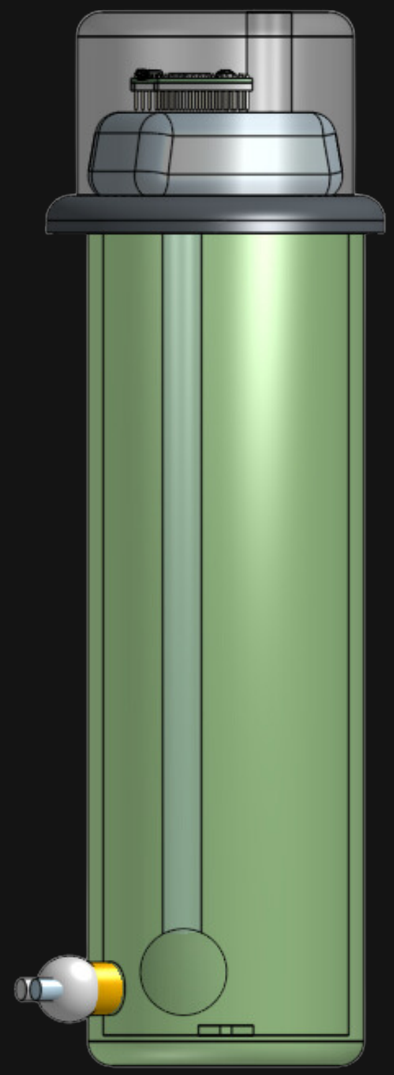


Valve B
Opened

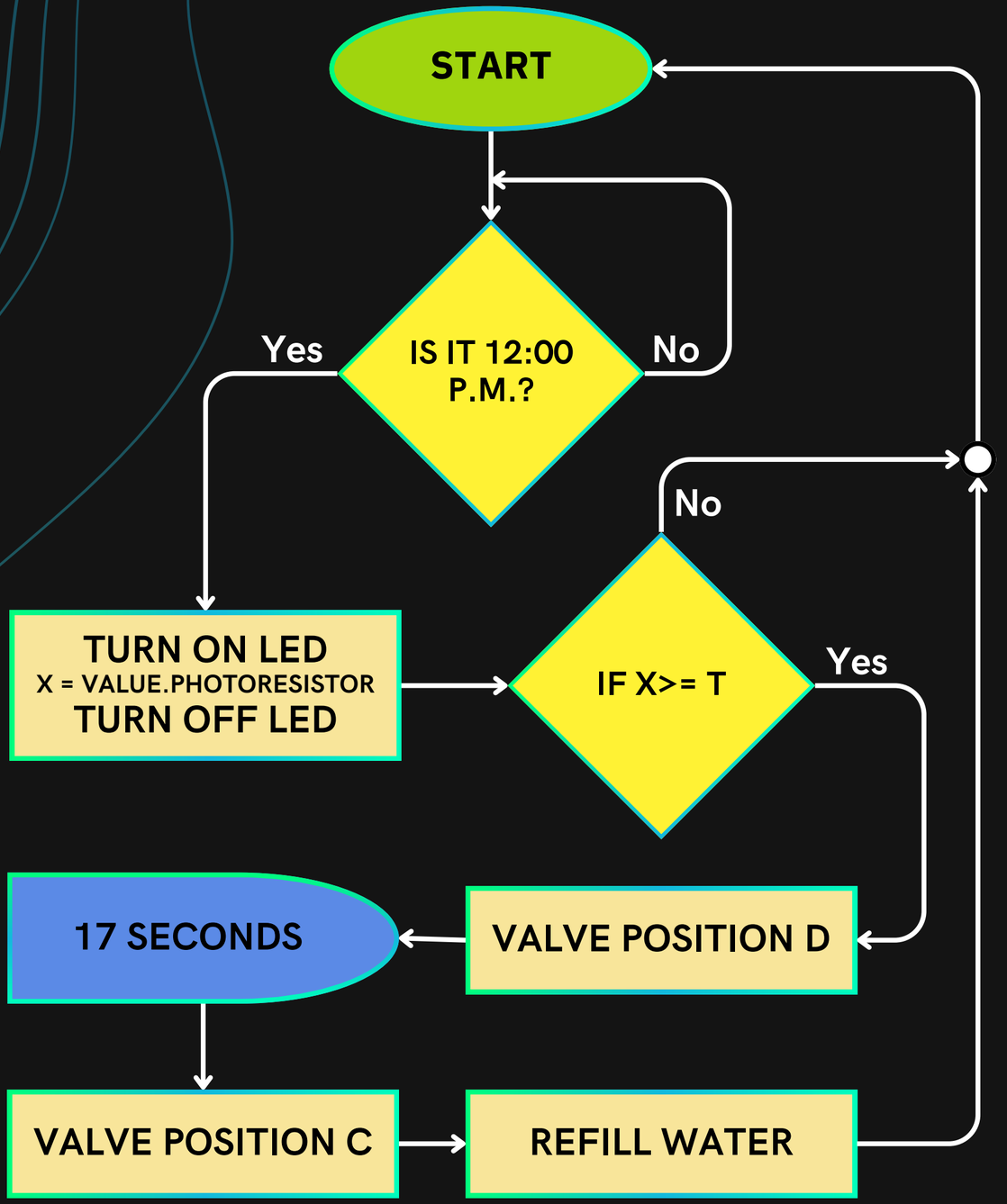
17 seconds

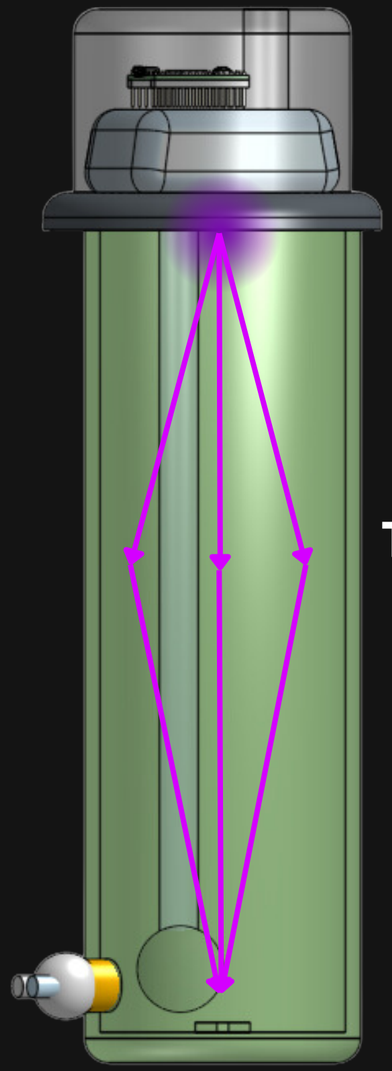
1 week cycle



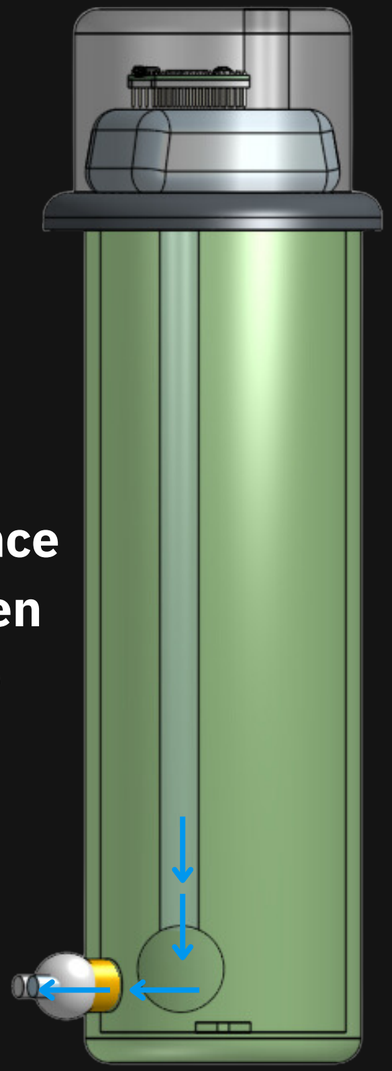
Valve B
Closed

# Algae Harvesting



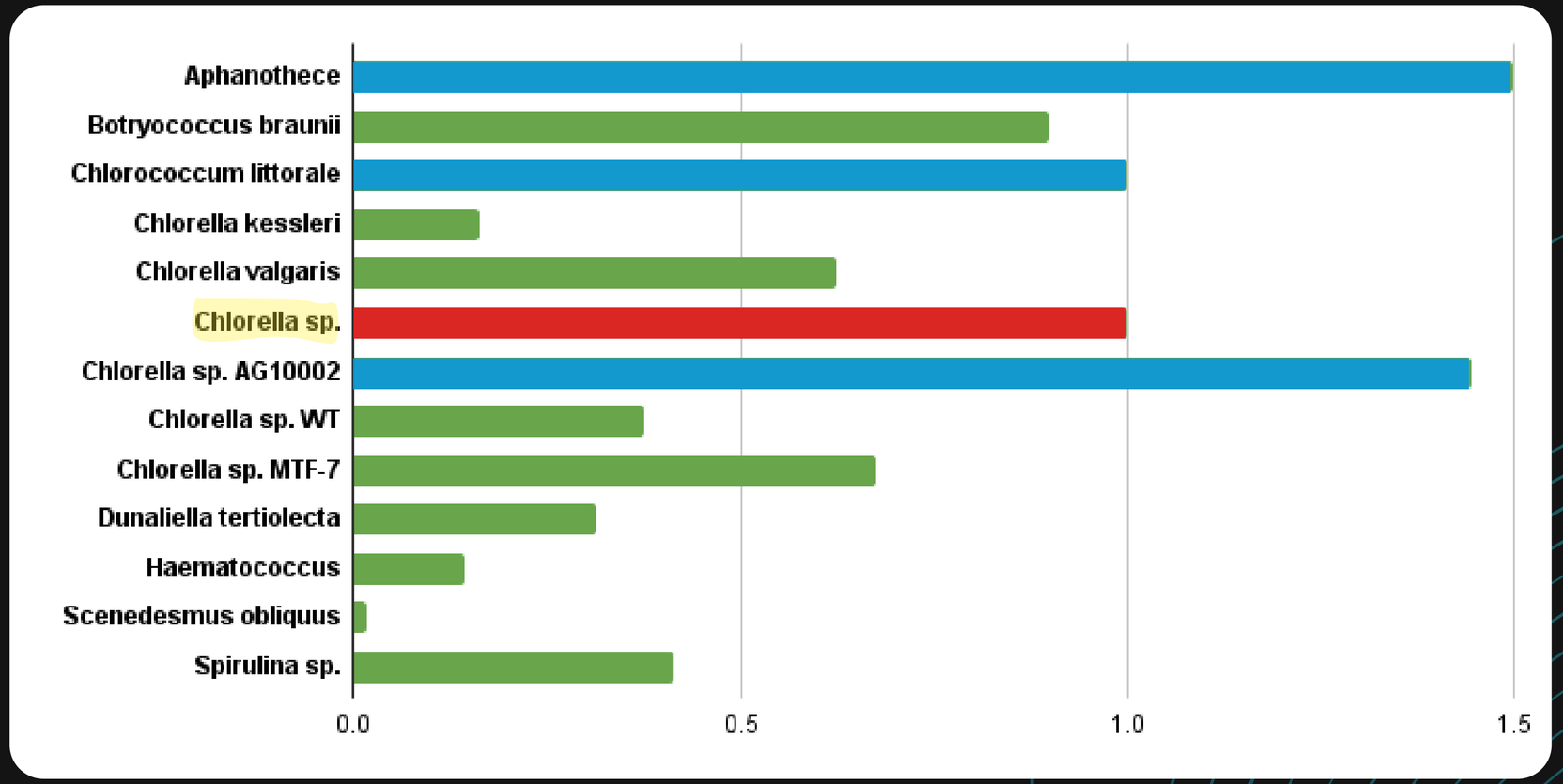



If the Transmittance exceed open the valve



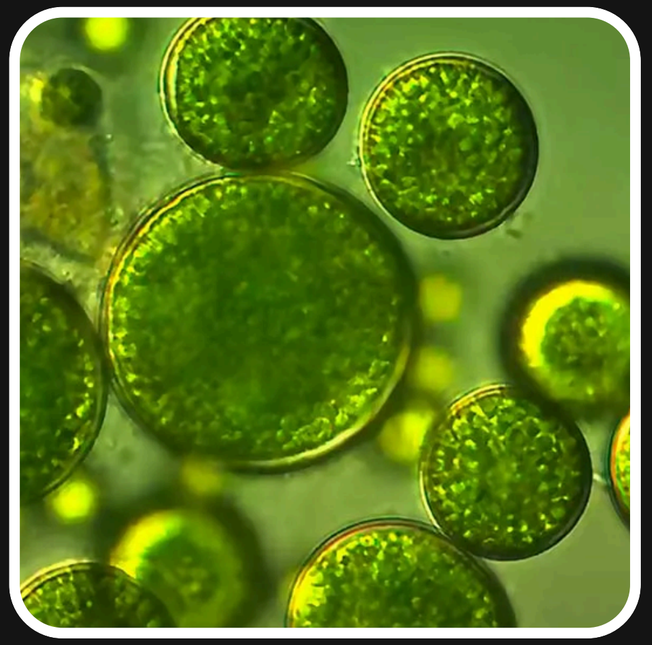
Valve A Opened

# Algae




Klinthong, W., Yang, Y., Huang, C., & Tan, C. (2015). A review: Microalgae and their applications in CO2 capture and renewable energy. Aerosol and Air Quality Research, 15(2), 712–742. https://doi.org/10.4209/aaqr.2014.11.0299

# Algae



GREEN ALGAE

*CHLORELLA SP.*

(CIPHERSKINCARE.COM, 2022)

"Chlorella is a specific strain of algae that is claimed to soak up much more $CO_2$ than any other plant with maximum rate of absorption of $CO_2$ = 1.0 g/L·day"

Politaeva, N., Ilin, I., Velmozhina, K., & Shinkevich, P. (2023). Carbon dioxide utilization using chlorella microalgae. Environments, 10(7), 109. https://doi.org/10.3390/environments10070109

# Decomposer

**BENEFICIAL BACTERIA**

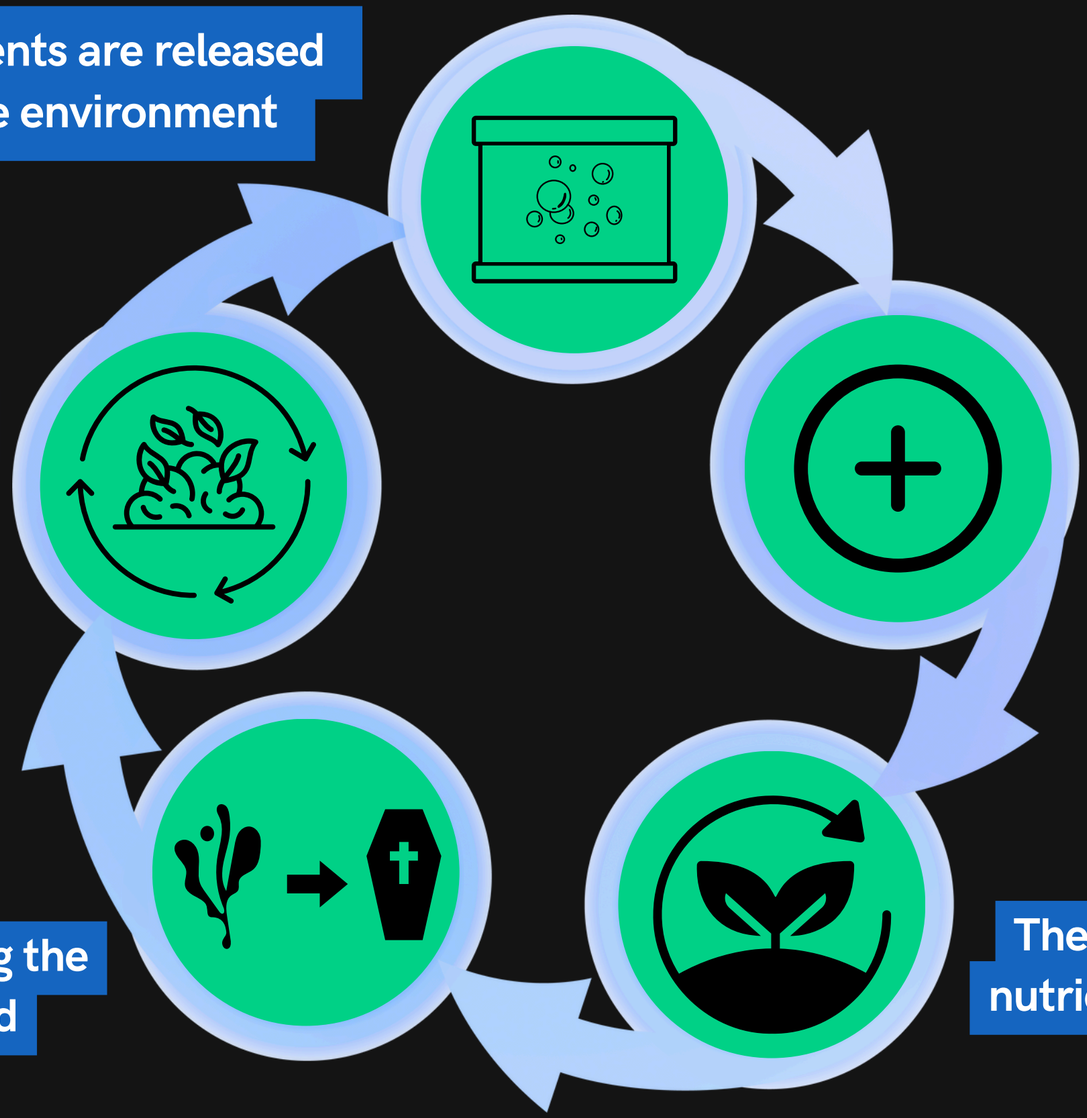
*BACILLUS SUBTILIS*

(BIONITY.COM, 2023)

"The Gram-positive bacterium Bacillus subtilis produces a variety of extracellular enzymes including six different proteases, an **α-amylase**, a **levansucrase**, several **β-glucanases** and at least two different **lipolytic enzymes [5–8]**"

Eggert, T., Brockmeier, U., Dröge, M. J., Quax, W. J., & Jaeger, K. (2003). Extracellular lipases fromBacillus subtilis: regulation of gene expression and enzyme activity by amino acid supply and external pH. FEMS Microbiology Letters, 225(2), 319–324. https://doi.org/10.1016/s0378-1097(03)00536-6

# Nutrient Cycle



# Price

| | Cost (Baht) |
|---|---|
| • STM32 Microcontroller | 54 |
| • Water pump (Small) | 38 |
| • Servo | 25 |
| • PLA Filament (40 grams) | 14 |
| • Photoresistor | 7 |
| • Air Stone | 7 |
| • Algae fertilizer (20mL) | 3 |
| • LED Light | 3 |
| • Rubber Tube | 2 |
| • Valve (3D print) | 0 |
| • Algae | 0 |

Total

153

Baht

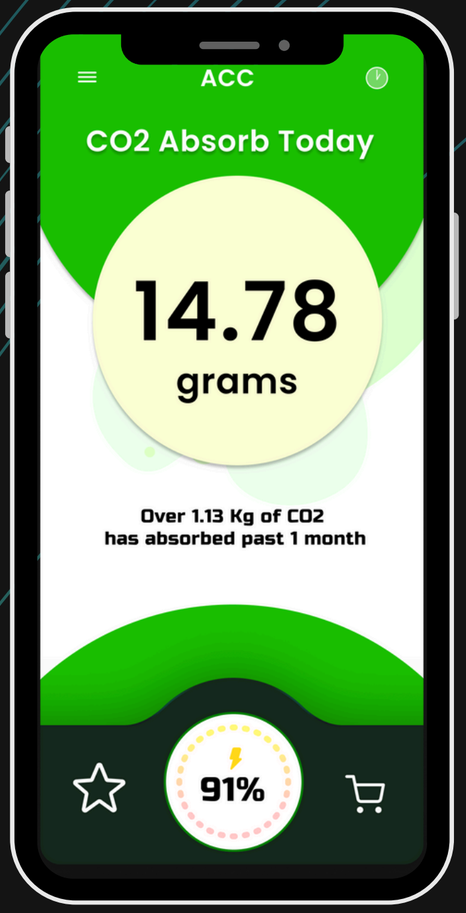
# Algae Waste

Livestock food supply

Fertilizer

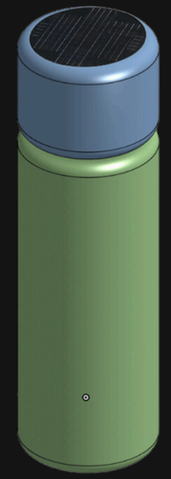
Biomass Fuel

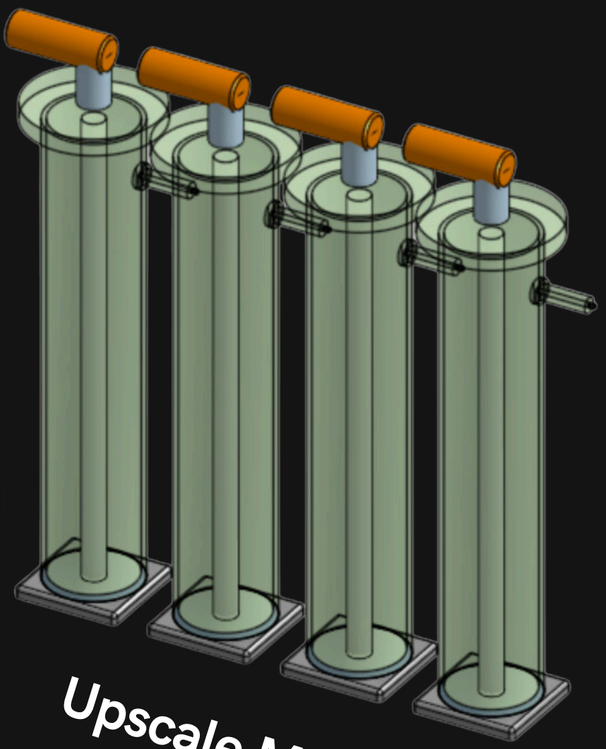
# Futher Improvement




Sample UX/UI

- Experiment in Lab.
- Make a real prototype
- Upscale the device
- Develop a Application that connect with ESP32 for real-time analysis



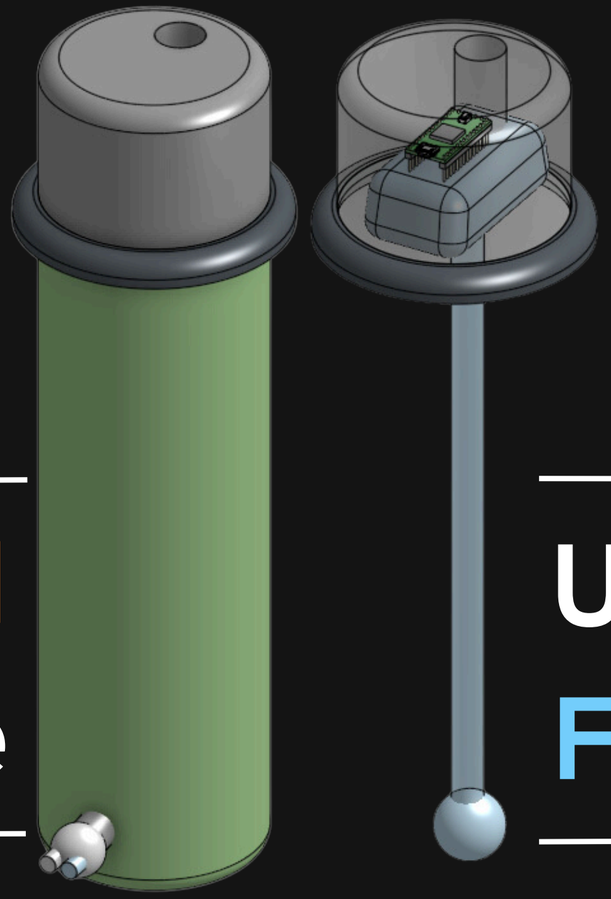


Upscale Model01

# Algal Carbon Capture

Absorbed $CO_2$

**16 kg** annually



Affordable price

only **฿153**

**Automated**

device

User

**Friendly**

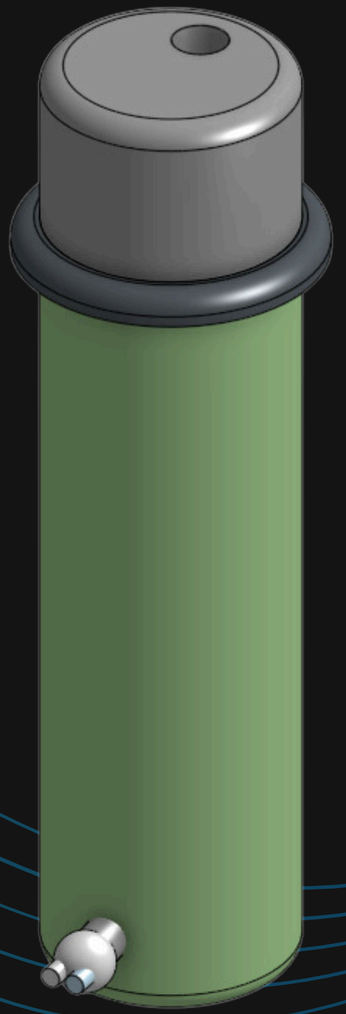
# References


- Stephenson, N., Das, A., Condit, R. et al. Rate of tree carbon accumulation increases continuously with tree size. Nature 507, 90–93 (2014). https://doi.org/10.1038/nature12914
- Iglina, T., Iglin, P., & Pashchenko, D. (2022). Industrial CO2 capture by algae: A review and recent advances. Sustainability, 14(7), 3801. https://doi.org/10.3390/su14073801
- Politaeva, N., Ilin, I., Velmozhina, K., & Shinkevich, P. (2023). Carbon dioxide utilization using chlorella microalgae. Environments, 10(7), 109. https://doi.org/10.3390/environments10070109
- Instructables. (2017, November 8). Simple Algae Home CO2 Scrubber - Part 1. Instructables. https://www.instructables.com/Simple-Algae-Home-CO2-Scrubber-Part-1/

Thank you

# Thep Hackathon SS2

# Algal Carbon Capture : Bio-Based solution for solving Climate Change

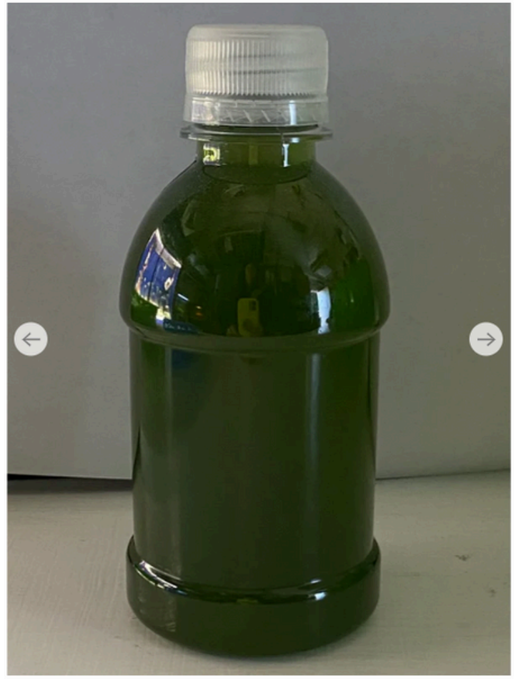


FINAL

MATTHEW

PEEMPONG

Home > Products > Algae Research Supply: Algae Culture Chlorella vulgaris



# ALGAE RESEARCH SUPPLY: ALGAE CULTURE CHLORELLA VULGARIS

Algae Research Supply

ACCv-00001

CULTURE VOLUME

1mL

QTY

1

ADD TO CART

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More payment options

297.00 ฿

*About the Algae: Chlorella is a genus of single-cell green algae belonging to the phylum Chlorophyta. It is spherical in shape, about 2 to 10 μm in diameter, and is without flagella. Chlorella contains the green photosynthetic pigments chlorophyll-a and -b in its chloroplast. Through photosynthesis, it multiplies rapidly, requiring only carbon dioxide, water, sunlight, and a small amount of minerals to reproduce.*

Forget genies, discover the magic of science with algae in a bottle! This product can be used to resupply your stash or continue the experimentation you began with our many kits and experiments. If you need more than just algae, consider our Spirulina Culture Kit which contains everything you need to grow a laboratory-grade culture at home or at your desk.

PLOT: Chlorella Secchi Disk Depth and Dry Weight as a Function of

Home > Pet Supplies > Fish & Aquatic > Aquarium & A... > Aq

# Sflora GO Algal 100ML + 20ML Algae Remov

1.8★ 5 Ratings & 1 Reviews

Special price

₹223 ~~₹240~~ 7% off

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View Details

Highlights — Solves Algae Problem

Seller — OxyIndia 4.1★

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# Micro Submersible Water Pump DC 3V-5V

★★★★★ 143 ระดับคะแนน

แบรนด์ China รหัสสินค้า: MO-SMWP-5V

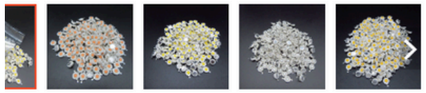
THB38.00 (VAT included)

เงินคืน: 3%

10 ชิ้นขึ้นไป THB35.50

สินค้าในสต็อก: 40

- 1 + เพิ่มในรถเข็น



ชิปหลอดไฟสปอตไลท์ LED COB 1W 3W Mini SMD RGB 440 660nm สีเขียว สีฟ้า สำหรับไฟดาวน์ไลท์ LED DIY 100 ชิ้น

ยังไม่มีคะแนน | 0 ขายแล้ว

~~฿394~~ ฿374 5% ส่วนลด

การจัดส่ง — ฟรีค่าจัดส่ง

การจัดส่ง จาก ต่างประเทศ ถึง เขตพระนคร, จังหวัดกรุงเทพมหานคร

ค่าจัดส่ง ฿0

สี

- สีขาวนวล 30000K
- สีแดง 625nm
- สีเขียว 520nm
- สีฟ้า 460nm
- รอยัลบลู 445nm
- สีเหลือง 2000K
- สเปกตรัมเต็มสีชมพู
- สีขาว สเปกตรัมเต็มรูปแบบ
- สีเหลือง 590-595nm
- สีเหลืองอำพัน 595-600nm
- สีส้ม 600-605nm
- 4pin RGB
- 6pin RGB
- Uv 395-400nm
- Uv420-430nm
- สีฟ้า 490nm

ขนาด

- 1W
- 3W

0

Aaairspecial Aquarium bubble air stone aerator ตู้ ปลาบ อบ

5.0 ★★★★★ 1

**คุณสมบัติ: ใหม่เอี่ยมและคุณภาพสูง 100% เพิ่มระดับ CO2 ที่ต่ำกว่าและเพิ่ม PH เคร**
**ปลาน้ำให้สำเร็จเพื่อกระบวนการช่วยหายใจ ทนทานล้างทำความสะอาดได้และสังกะสี**

ดูรายละเอียดผลิตภัณฑ์

**฿7.00**

Shopee

+ ค่าจัดส่ง

แชร์: Favorite (27)

หัวเชื้อน้ำเขียวคลอเรลล่าน้ำจืด (Chlorella) คุณภาพผลิตจาก
3 ขวด (ขวดละ 500 ml.)

4.8 ★★★★★ | 58 Ratings | 149 ขายแล้ว

**฿220**

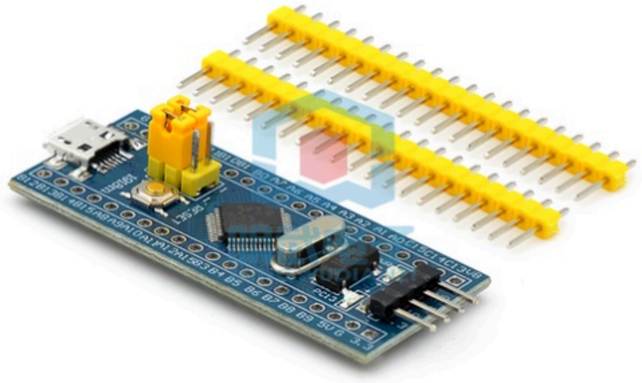
การจัดส่ง: การจัดส่ง ถึง เขตพระนคร, จังหวัดกรุงเทพมหานค
ค่าจัดส่ง ฿64

จำนวน: − 1 + มีสินค้าทั้งหมด 0 ชิ้น

เพิ่มไปยังรถเข็น | ซื้อสินค้า

ช้อปปี้การันตี รับประกันคืนเงิน หากไม่ได้รับสินค้า

型号：STM32F103C6T6
尺寸：53mm*22.6mm



ร้านแนะนำ [ราคาที่ดีสำหรับปริมาณมาก] STM32F103C8T6 /
MCU Core Board STM32 ARM

ยังไม่มีคะแนน | 10 ขายแล้ว

**฿54**

ช้อปเพิ่มคุ้มกว่า: ช้อปเพิ่มคุ้มกว่า

การจัดส่ง: ฟรีค่าจัดส่ง
รับส่วนลดค่าจัดส่ง เมื่อสั่งซื้อถึง ฿499

การจัดส่ง จาก: ต่างประเทศ ถึง เขตพระนคร, จังหวัดกรุงเทพมหานคร

ค่าจัดส่ง: ฿0 - ฿23

การจำแนกสี: 103C8T6开发板 国产ST | STM32F103C6T6 开发板 | STM32F030C8T6

หัวเชื้อน้ำเขียว chlorella+
ประกอบไปด้วยหัวเชื้อน้ำเขียว 3 ขวด ๆ ละ 500 มิลลิตร

คลอเรลลา (Chlorella sp.) เป็นสาหร่ายสีเขียวเซลล์เดียวขนาดเล็ก มีขนาด 2.5 - 3.5 ไมครอน มีโปรตีนสูง 64.15% เหมาะที่จะนำไปเพาะขยายเพื่อเป็นอาหารเลี้ยงไรแดง

การเพาะขยายคลอเรลลาสามารถทำได้ไม่ยากใช้เวลาเพียง 3-4 วัน ก็สามารถได้น้ำที่มีสีเขียวเข้มเพื่อเป็นอาหารของไรแดง

วิธีการเพาะขยายน้ำเขียวคลอเรลลาโดยการใช้ปุ๋ยอนินทรีย์(ปุ๋ยเคมี อาทิเช่น ปุ๋ยยูเรีย, ปุ๋ยนา, ปุ๋ยสูตรเสมอ) ร่วมกับปุ๋ยอินทรีย์หรืออินทรีย์วัตถุต่างๆเช่น ปลาป่น, รำละเอียด, กากผงชูรส) ใช้ระยะเวลา
การเพาะขยายเพื่อทำให้น้ำเขียวเข้มประมาณ 3-4 วัน

นอกจากนั้นหัวเชื้อน้ำเขียวคลอเรลลายังสามารถนำไปใส่ในบ่อปลากลางแจ้ง
- ช่วยลดความเข้มของแสงที่ส่องลงไปในบ่อ
- ช่วยลดความเป็นพิษของเลียในบ่อ
- ช่วยเพิ่มออกซิเจนและลดก๊าซคาร์บอนไดออกไซด์ให้กับบ่อเลี้ยงสัตว์น้ำ
- ช่วยลดความเครียดให้ปลาและสัตว์น้ำในบ่อ

*หมายเหตุ: ชุดนี้เหมาะสำหรับผู้ที่มีปุ๋ยแล้วค่ะ

| | ABS | Flexible | PLA | HIPS | PETG | Nylon | Carbon Fiber Filled | ASA |
|---|---|---|---|---|---|---|---|---|
| | Learn More | Learn More | Learn More | Learn More | Learn More | Learn More | Learn More | Learn More |
| Compare Selected | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Ultimate Strength | 40 MPa | 26 - 43 MPa | 65 MPa | 32 MPa | 53 MPa | 40 - 85 MPa | 45 - 48 MPa | 55 MPa |
| Stiffness | 5 / 10 | 1 / 10 | 7.5 / 10 | 10 / 10 | 5 / 10 | 5 / 10 | 10 / 10 | 5 / 10 |
| Durability | 8 / 10 | 9 / 10 | 4 / 10 | 7 / 10 | 8 / 10 | 10 / 10 | 3 / 10 | 10 / 10 |
| Maximum Service Temperature | 98 °C | 60 - 74 °C | 52 °C | 100 °C | 73 °C | 80 - 95 °C | 52 °C | 95 °C |
| Coefficient of Thermal Expansion | 90 µm/m-°C | 157 µm/m-°C | 68 µm/m-°C | 80 µm/m-°C | 60 µm/m-°C | 95 µm/m-°C | 57.5 µm/m-°C | 98 µm/m-°C |
| Density | 1.04 g/cm³ | 1.19 - 1.23 g/cm³ | 1.24 g/cm³ | 1.03 - 1.04 g/cm³ | 1.23 g/cm³ | 1.06 - 1.14 g/cm³ | 1.3 g/cm³ | 1.07 g/cm³ |
| Price (per kg) | $10 - $40 | $30 - $70 | $10 - $40 | $24 - $32 | $20 - $60 | $25 - $65 | $30 - $80 | $38 - $40 |
| Printability | 8 / 10 | 6 / 10 | 9 / 10 | 6 / 10 | 9 / 10 | 8 / 10 | 8 / 10 | 7 / 10 |
| Extruder Temperature | 220 - 250 °C | 225 - 245 °C | 190 - 220 °C | 230 - 245 °C | 230 - 250 °C | 220 - 270 °C | 200 - 230 °C | 235 - 255 °C |

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = constant$$

ตอบ

$$P_1 + \rho g h_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho g h_2 + \frac{1}{2} v_2^2$$

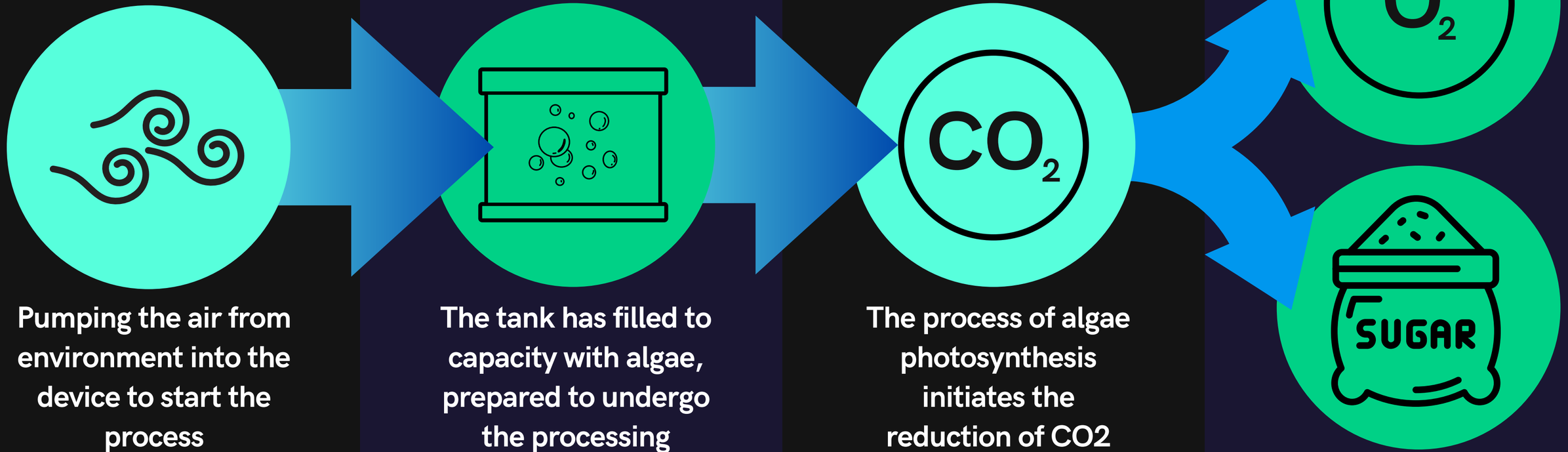
$$\rightarrow v_2 = \sqrt{2gh_2}$$

$$= 2.21 \text{ m/s}$$

ตอบ $Q = AV = \frac{V}{t}$

$$\therefore t \sim 17 \text{ s.}$$

# Process




Pumping the air from environment into the device to start the process

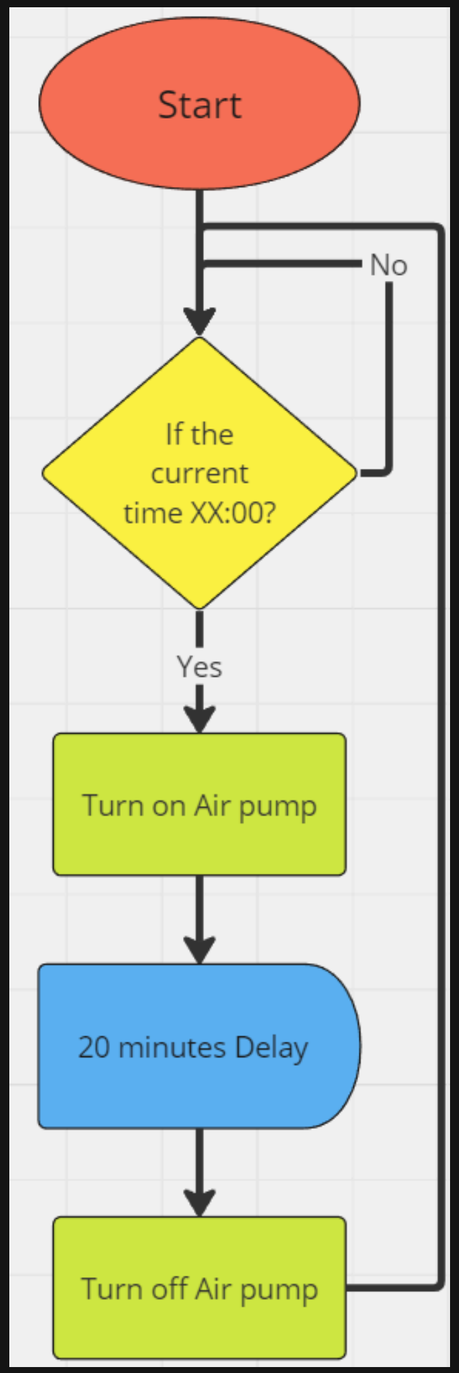
The tank has filled to capacity with algae, prepared to undergo the processing procedure.

The process of algae photosynthesis initiates the reduction of CO2

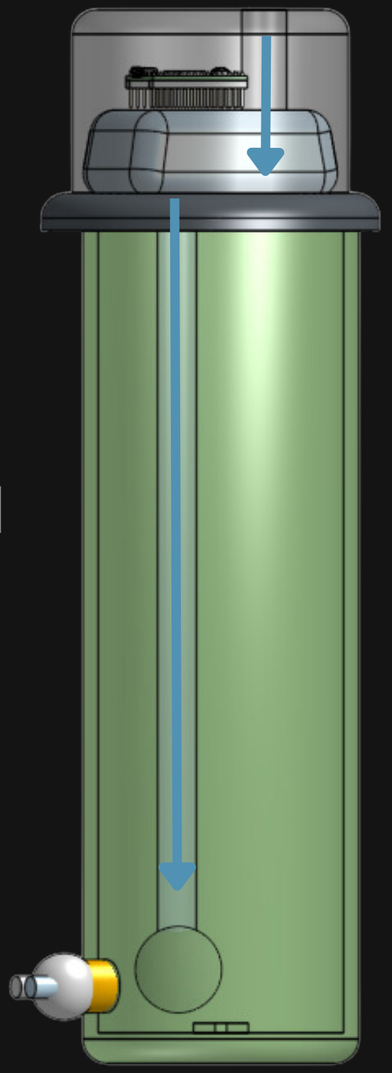
Release O2 into the environment to end the process

Carbon dioxide (CO2) is utilized by the algae to produce glucose

# Air Pump Flow Chart

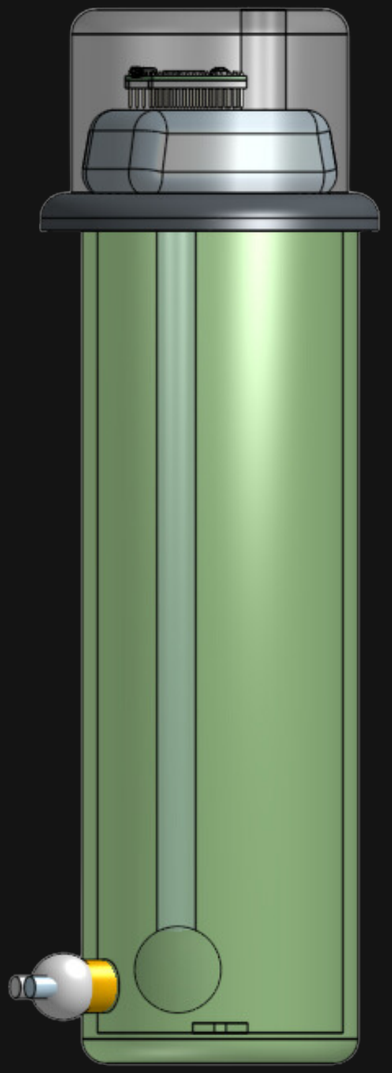



**Pump Activated**



**20 Minutes**

**1 hour cycle**



**Pump Inactivated**